

AIDOS LAB
AI FOR DATA-ORIENTED SCIENCE

UNI FR

Geometric and topological methods in deep learning

XIV Encuentro de Jóvenes Topólogos — Logroño, Universidad de La Rioja

Inés García Redondo — 21/10/2026

Session 2 — Topological losses and generalization

Outline

Topological Deep Learning

Learning on topological domains

How can networks process higher-order relational data?

Incorporating topology into learning

How can topological information guide a model?

Understanding networks through topology

What can shape reveal about learning and model behaviour?

Topological losses:

PersLay

Learning filtrations

Topological Autoencoders

Generalization and PH:

Neural persistence

Predicting the generalization gap using PH of activations and PH dimension

PersLay: learning from persistence diagrams

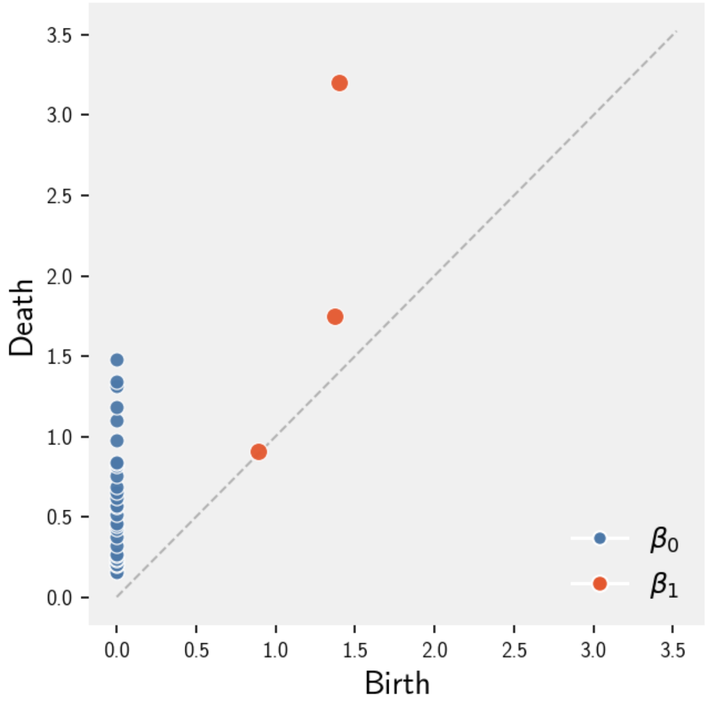
Session 2 — Geometric and topological methods in deep learning

Persistent homology

The general pipeline

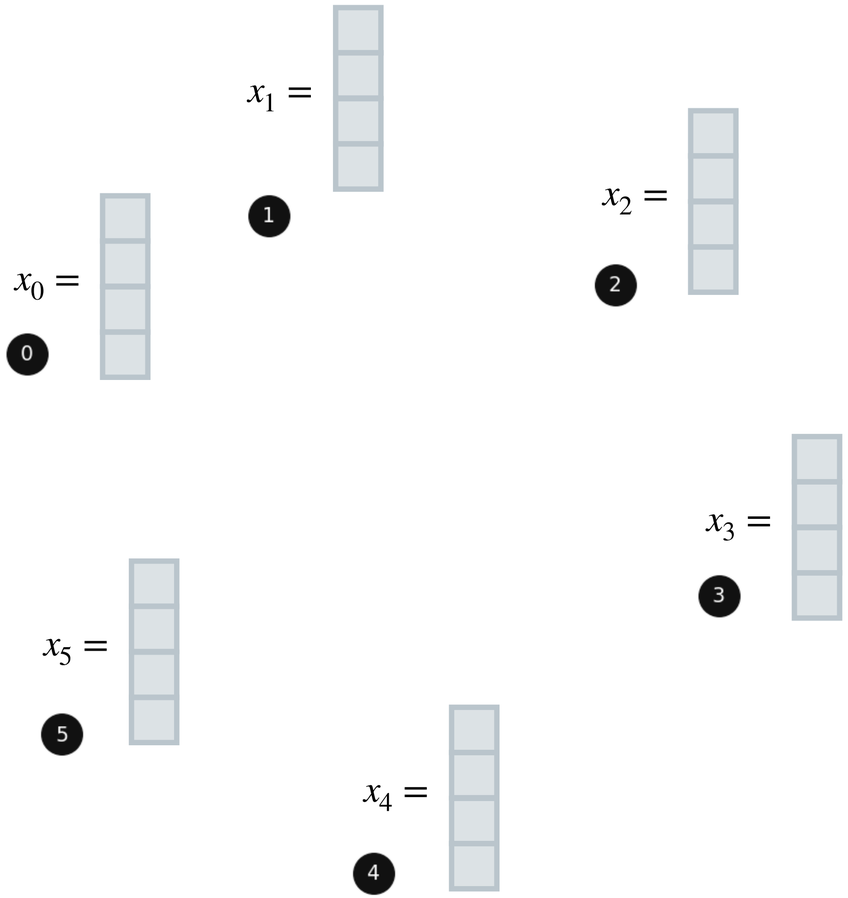


Point cloud ⟶ Čech ⟶ Persistence diagram ⟶ ???



Learning from sets

The DeepSet architecture



- Suppose we are given a set of points S, and we want to learn a function $f : S \to \mathbb{R}^k$
- In other words, we want to learn feature vectors on the points of the set: $x_i \in \mathbb{R}^d$ for $i \in S$
- We stack them: $X = (x_1, \ldots, x_n)^\top \in \mathbb{R}^{n \times d}$

  Already specifies an order in which you are visiting the nodes

  But our function should not depend on this choice!

What transformations change the ordering of the nodes?

Permutations $\sigma \in S_n$

Can be represented as *permutation matrices*, which we can apply to X

Example:

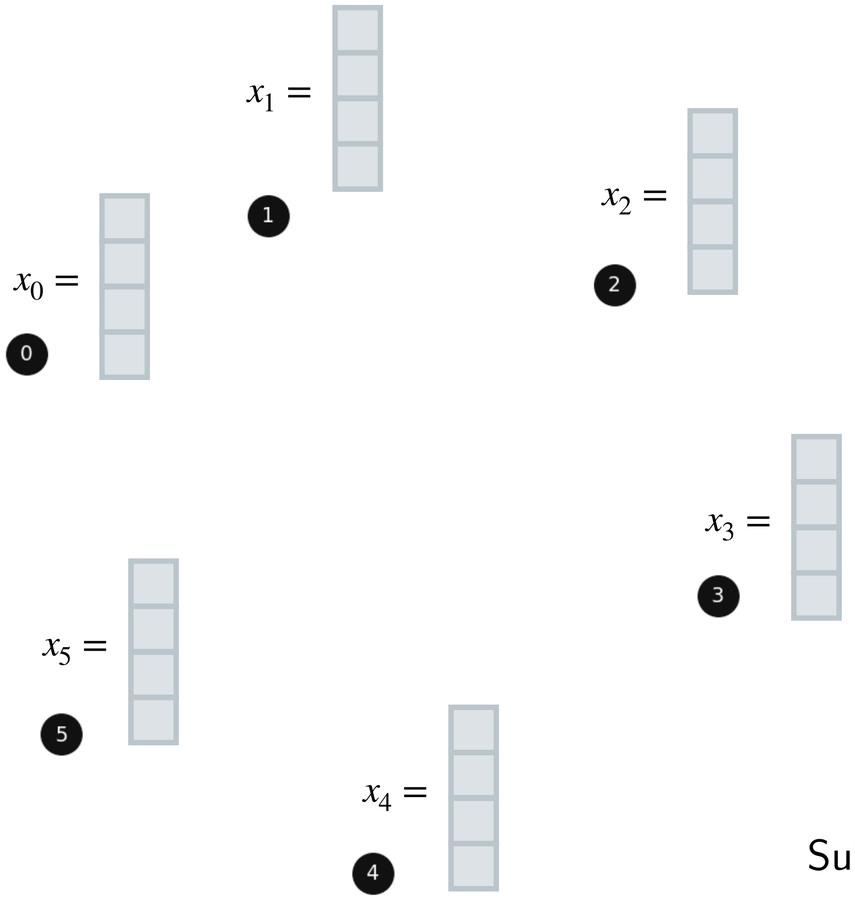
$$\sigma = (1\,3\,2\,4) \in S_4$$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix}$$

$$P_\sigma = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \; X = \begin{pmatrix} x_1^\top \\ x_2^\top \\ x_3^\top \\ x_4^\top \end{pmatrix}, \quad P_\sigma X = \begin{pmatrix} x_3^\top \\ x_4^\top \\ x_2^\top \\ x_1^\top \end{pmatrix}$$

Learning from sets

The DeepSet architecture



Goal: define a function $f : \mathbb{R}^{n\times d} \to \mathbb{R}^k$ taking as input

$$X = (x_1, \dots, x_n)^\top \in \mathbb{R}^{n\times d}$$

which is *permutation* invariant, that is, which for all permutation matrices P

$$f(PX) = f(X)$$

Theorem

Any such function can be written as

$$f(X) = \varphi\left(\sum_{i\in S} \phi(\mathbf{x}_i)\right)$$

for suitable $\phi : \mathbb{R}^d \to \mathbb{R}^m, \quad \varphi : \mathbb{R}^m \to \mathbb{R}^k$

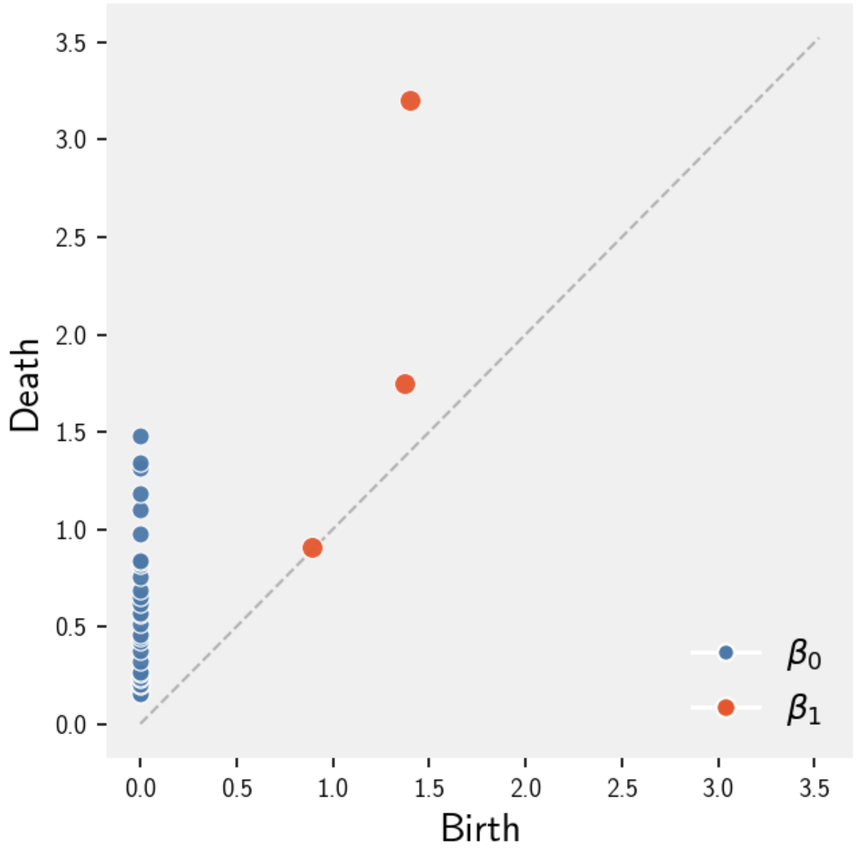
DeepSets are the architecture built in this manner, to learn from sets

Substituting the sum by any other permutation invariant operator like taking the maximum, the kth maximum, or an average, you still obtain a permutation invariant function

[1]Zaheer, M., Kottur, S., Ravanbakhsh, S., Poczos, B., Salakhutdinov, R. R., and Smola, A. (2017). Deep Sets. *Advances in Neural Information Processing Systems*, *30*.

Learning from persistence diagrams

PersLay



Extend the DeepSet architecture to accommodate persistence diagrams

$$\mathrm{PersLay}(\mathrm{PD}) := \bigoplus_{p \in \mathrm{PD}} w_{\theta_1}(p) \cdot \varphi_{\theta_2}(p)$$

With $\oplus$ a permutation invariant operator, $w_{\theta_1} : \mathbb{R}^2 \to \mathbb{R}$ a weight function for the points in the diagram and $\varphi_{\theta_2} : \mathbb{R}^2 \to \mathbb{R}^q$ a *point transformation* (a vectorization)

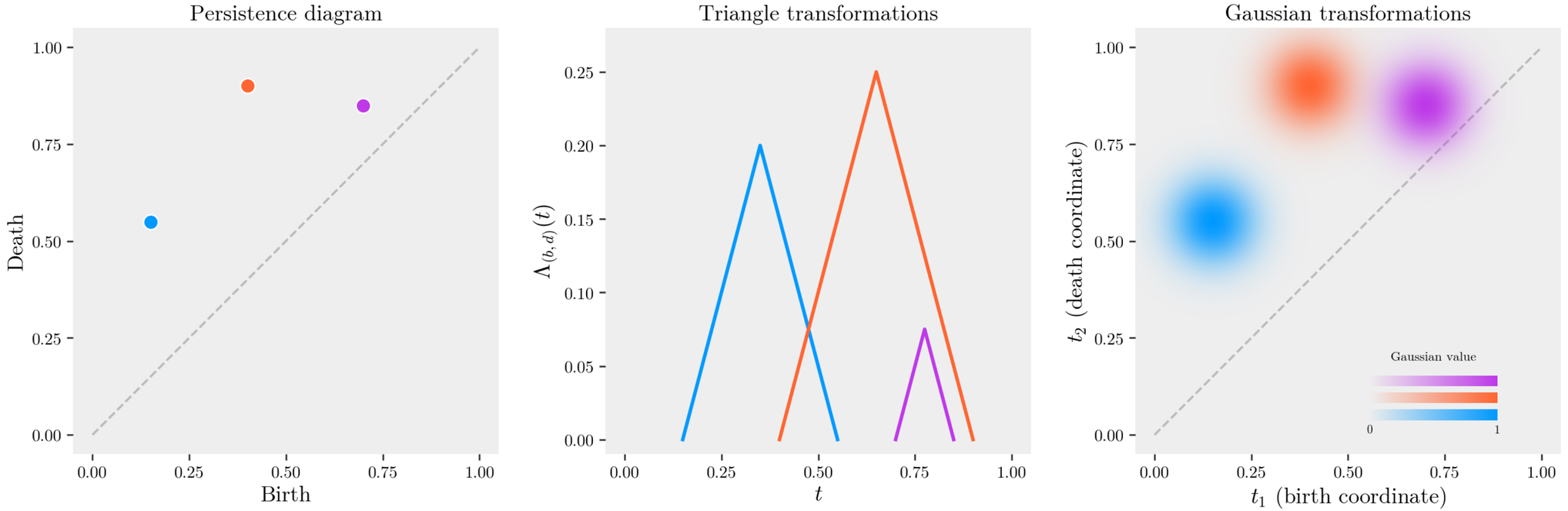
This general formulation allows to recover several popular vectorizations

Carrière, M., Chazal, F., Ike, Y., Lacombe, T., Royer, M., and Umeda, Y. (2020). PersLay: A Neural Network Layer for Persistence Diagrams and New Graph Topological Signatures. *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, 2020.

Learning from persistence diagrams

Point transformations

- **Triangle point transformation:** $\phi_\Lambda : \mathbb{R}^2 \to \mathbb{R}^q$, $p = (x, y) \mapsto \big(\Lambda_p(t_1), \ldots, \Lambda_p(t_q)\big)$ where $\Lambda_p : \mathbb{R} \to \mathbb{R}$, $t \mapsto \max\{0, y - |t - x|\}$, for $q \in \mathbb{N}$, $t_1, \ldots, t_q \in \mathbb{R}$ (learnable parameters)
- **Gaussian point transformation:** $\phi_\Gamma : \mathbb{R}^2 \to \mathbb{R}^q$, $p = (x, y) \mapsto \big(\Gamma_p(t_1), \ldots, \Gamma_p(t_q)\big)$ where $\Gamma_p : \mathbb{R}^2 \to \mathbb{R}$, $t \mapsto \exp\left(\frac{-\|p - t\|^2}{2\sigma^2}\right)$, for $\sigma > 0$ (learnable parameter) and $q \in \mathbb{N}$, $t_1, \ldots, t_q \in \mathbb{R}^2$ (fixed)

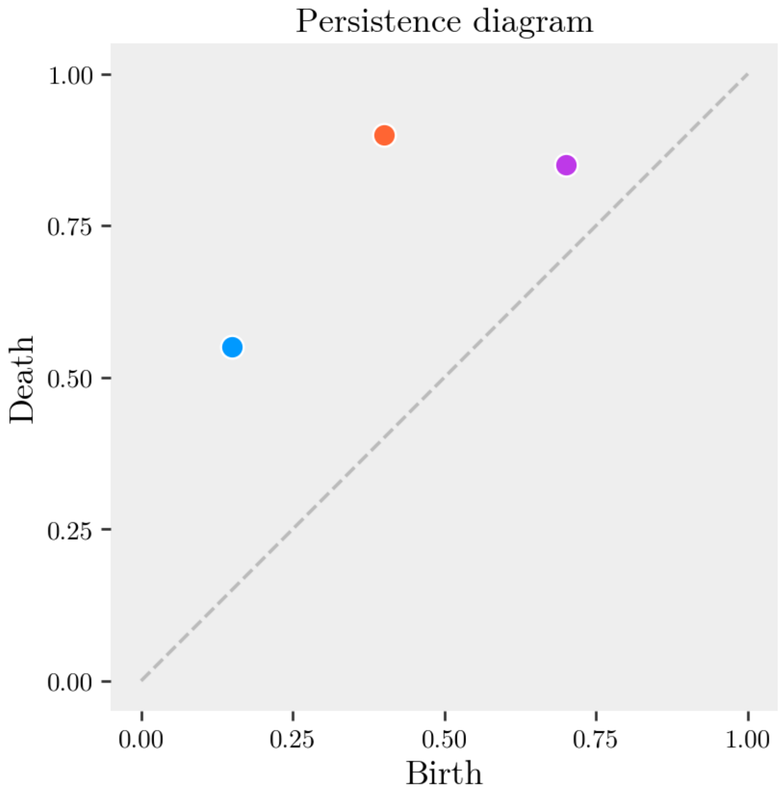


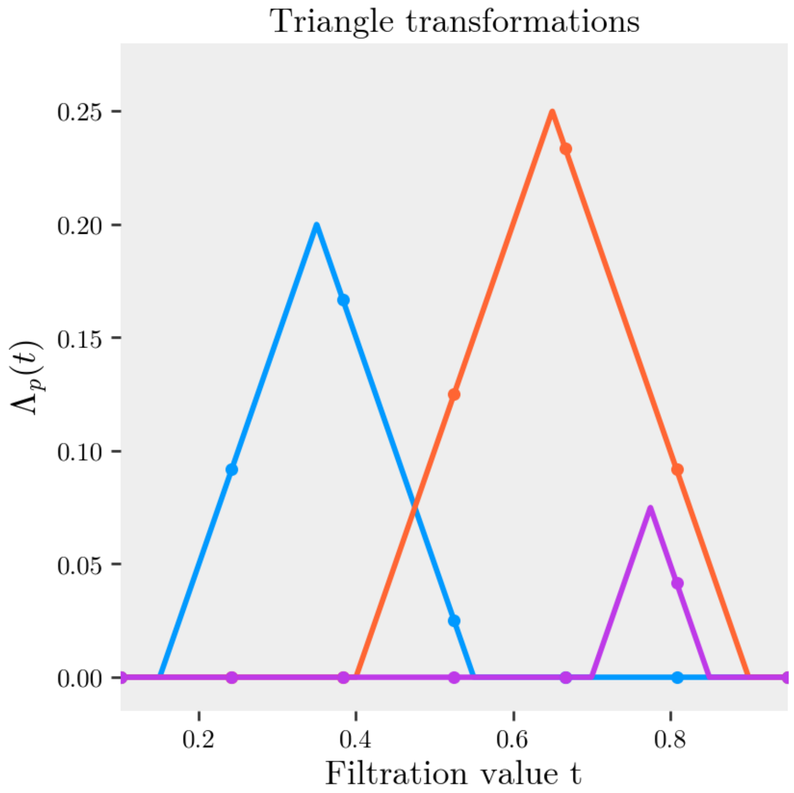
Learning from persistence diagrams

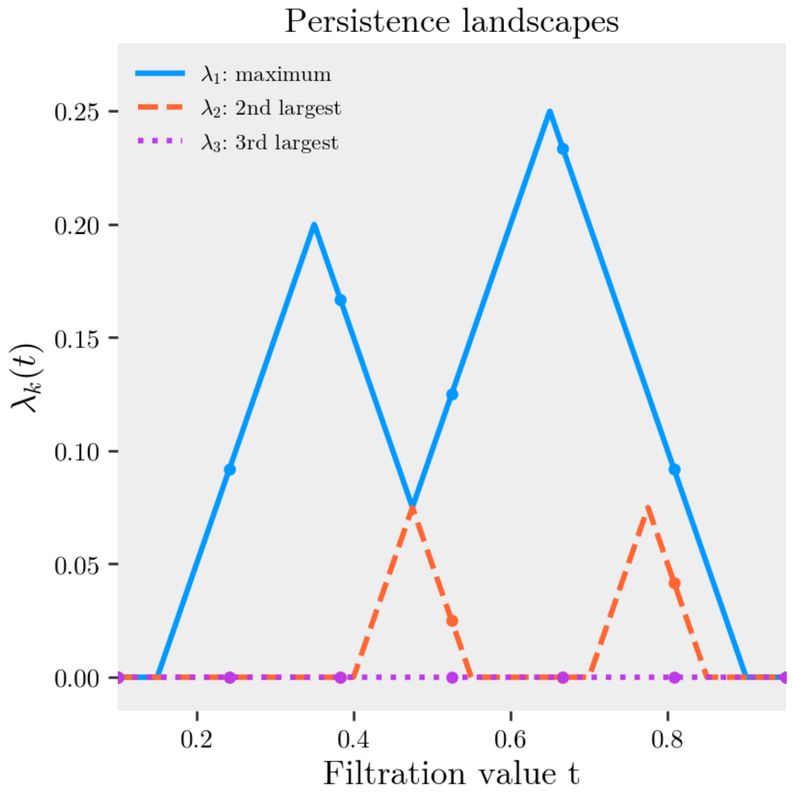
PersLay recovers landscapes

$$\phi = \phi_\Lambda \text{ (triangle point transformation)}, \quad w(p) = 1, \forall p, \quad \bigoplus = k\text{-th largest value per coordinate}$$

$$\left(\operatorname*{kmax}_{p \in \mathrm{PD}} \Lambda_p(t_1), \ldots, \operatorname*{kmax}_{p \in \mathrm{PD}} \Lambda_p(t_q) \right)\}$$







Application to graph classification

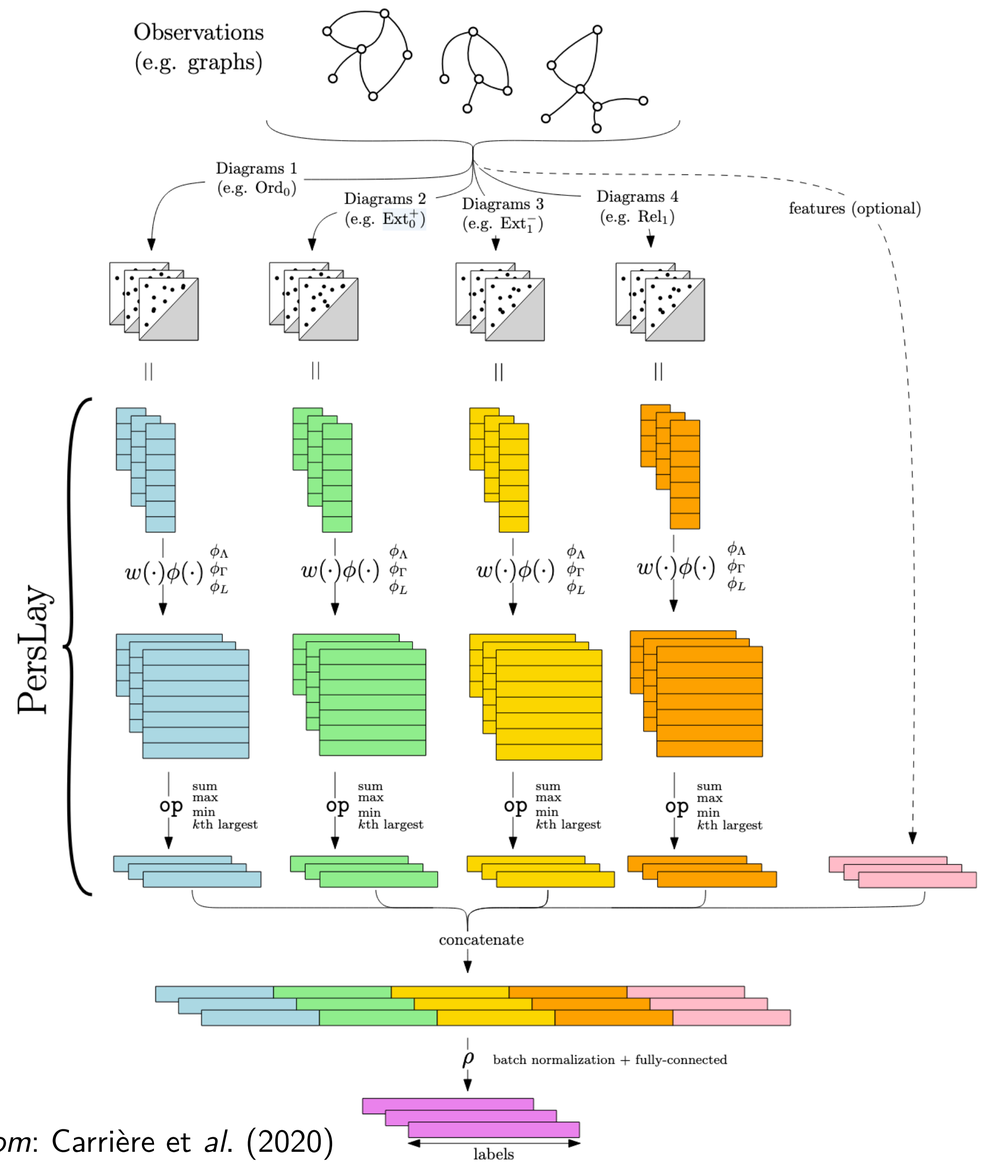


Figure from: Carrière et *al.* (2020)

Input data: graphs, seen as discrete metric spaces with a distance defined on the vertices according to the graph structure

Compute **persistence diagrams** from *extended persistence*

Apply the `PersLay` transformation for three point transformations and learnable weights

Concatenate all resulting vectors + normalize

Apply an **MLP** to classify

Application to graph classification

Datasets studied

Dataset	Nb graphs	Nb classes	Av. nodes	Av. Edges	Av. β_0	Av. β_1
REDDIT5K	5,000	5	508.5	594.9	3.71	90.1
REDDIT12K	12,000	11	391.4	456.9	2.8	68.29
COLLAB	5,000	3	74.5	2457.5	1.0	2383.7
IMDB-B	1,000	2	19.77	96.53	1.0	77.76
IMDB-M	1,500	3	13.00	65.94	1.0	53.93
COX2	467	2	41.22	43.45	1.0	3.22
DHFR	756	2	42.43	44.54	1.0	3.12
MUTAG	188	2	17.93	19.79	1.0	2.86
PROTEINS	1,113	2	39.06	72.82	1.08	34.84
NCI1	4,110	2	29.87	32.30	1.19	3.62
NCI109	4,127	2	29.68	32.13	1.20	3.64

Social graphs

Molecular datasets

Table from: Carrière et *al.* (2020)

Application to graph classification

Results

Dataset	SV[1]	RetGK* [2]	FGSD [3]	GCNN [4]	GIN [5]	PersLay	
						Mean	Max
REDDIT5K	—	56.1	47.8	52.9	57.0	55.6	56.5
REDDIT12K	—	48.7	—	46.6	—	47.7	49.1
COLLAB	—	81.0	80.0	79.6	80.1	76.4	78.0
IMDB-B	72.9	71.9	73.6	73.1	74.3	71.2	72.6
IMDB-M	50.3	47.7	52.4	50.3	52.1	48.8	52.2
COX2*	78.4	80.1	—	—	—	80.9	81.6
DHFR*	78.4	81.5	—	—	—	80.3	80.9
MUTAG*	88.3	90.3	92.1	86.7	89.0	89.8	91.5
PROTEINS*	72.6	75.8	73.4	76.3	75.9	74.8	75.9
NCI1*	71.6	84.5	79.8	78.4	82.7	73.5	74.0
NCI109*	70.5	—	78.8	—	—	69.5	70.1

Table from: Carrière et *al.* (2020)

- SV: kernel method on persistence diagrams for point clouds obtained from the graphs
- RetBK: graph kernel method using graph attributes
- FGSD: graph kernel method that doesn't use graph attributes
- GCNN, GIN: graph neural networks

Limitations of PersLay

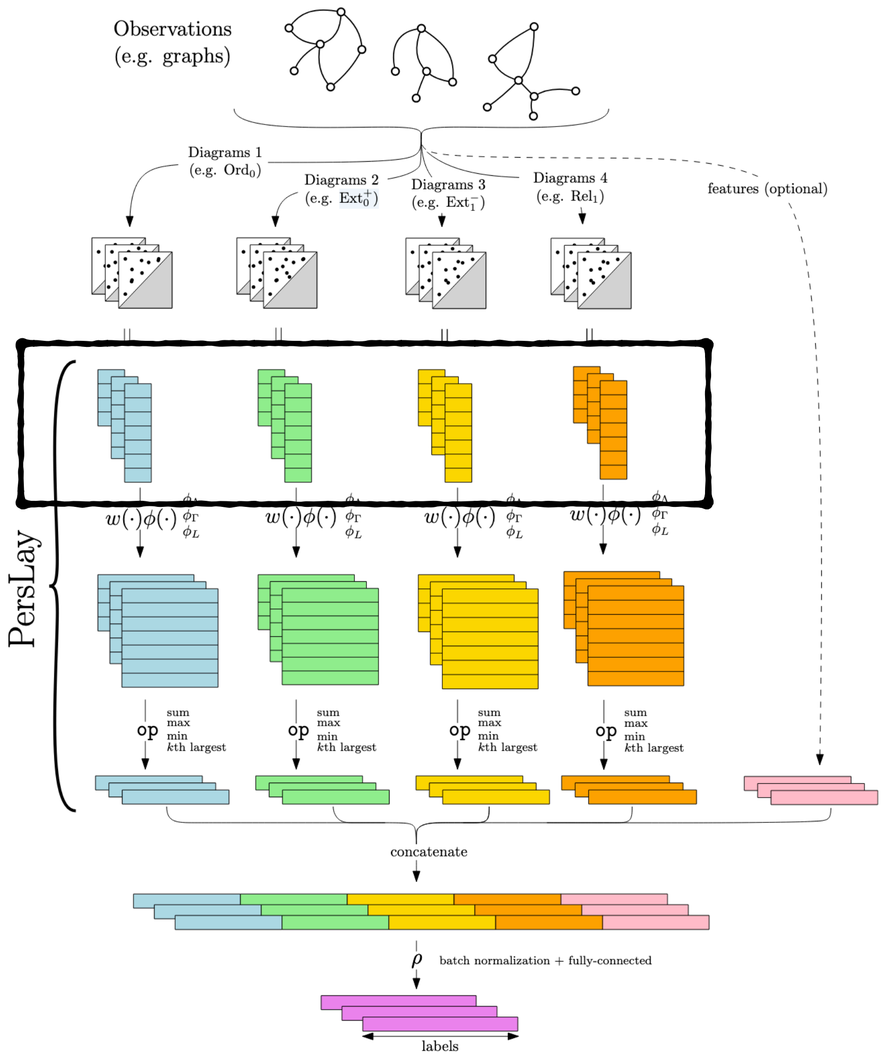


Figure from: Carrière et *al.* (2020)

Not learnable: the persistence computation is fixed, can we make it learnable?

Can we differentiate wrt any parameters here?

Learnable part: where the parameters that we are optimizing live

Topological losses: learning the filtration

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Defining a loss on the persistence diagram

Persistence diagram: $\mathrm{PD}_k = \{(b_i, d_i) : i \in \mathcal{I}_k\}$

Order points by decreasing lifetime $d_i - b_i \geq d_j - b_j \iff i < j$

Goal: define a *differentiable* loss of these diagrams

$$\mathcal{L}(p, q, i_0; \mathrm{PD}_k) = \sum_{i=i_0}^{|\mathcal{I}_k|} |d_i - b_i|^p \left(\frac{d_i + b_i}{2}\right)^q$$

Leave the first i_0 most persistent features out

Increasing p penalizes more the most persistent features

Weight that gives more importance to bars appearing later in the filtration

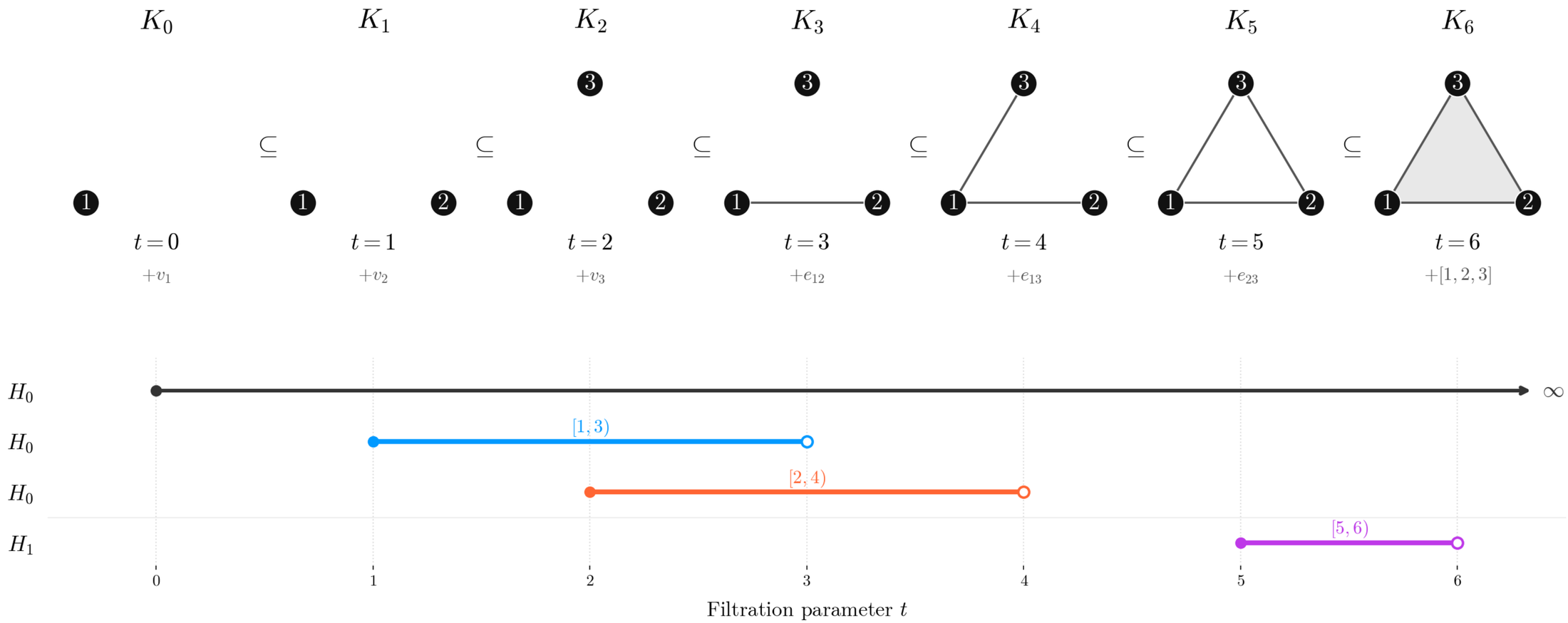
Can we link the coordinates of the persistence diagram to the filtration?

Gabrielsson, R. B., Nelson, B. J., Dwaraknath, A., and Skraba, P. (2020). A Topology Layer for Machine Learning. *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, 1553–1563.

Simplex-wise filtrations

Def. Simplex-wise filtration

A filtration $\{X_t : t \in \mathbb{R}\}$ such that at each parameter value $t \in \mathbb{R}$ at most one simplex enters the filtration



Gradient of the loss

Critical simplices: simplices that created and destroyed the topological feature represented by the bar

If the filtration is simplex-wise, we have a *unique map:*

$$\pi_k : \{(b_i, d_i) : i \in \mathcal{I}_k\} \to (\sigma, \tau)$$

Identify a simplex with the parameter value where it enters the filtration. We have that the *gradient* of a loss defined on the persistence diagram will be:

$$\frac{\partial \mathcal{L}}{\partial \sigma} = \sum_{i=0}^{|\mathcal{I}_k|} \frac{\partial \mathcal{L}}{\partial b_i} \mathbf{1}_{\pi_k(b_i)=\sigma} + \sum_{i=0}^{|\mathcal{I}_k|} \frac{\partial \mathcal{L}}{\partial d_i} \mathbf{1}_{\pi_k(d_i)=\sigma}$$

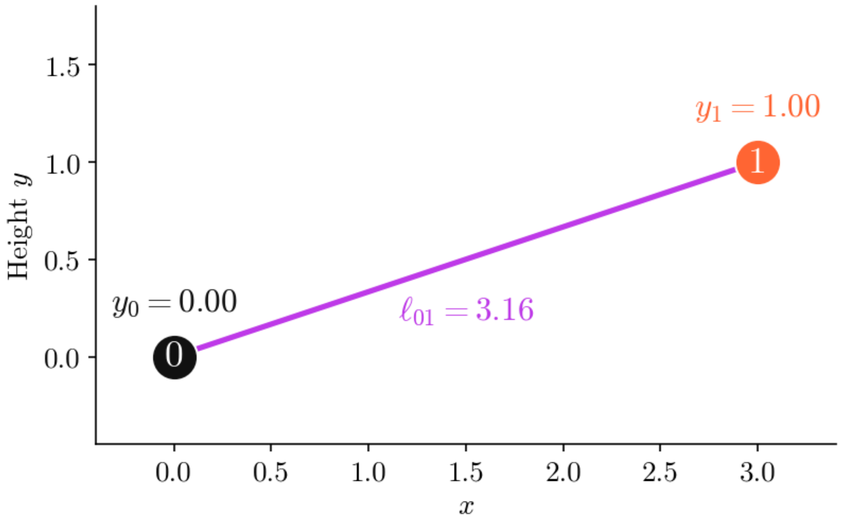
Where at most one term will have non-zero indicator.

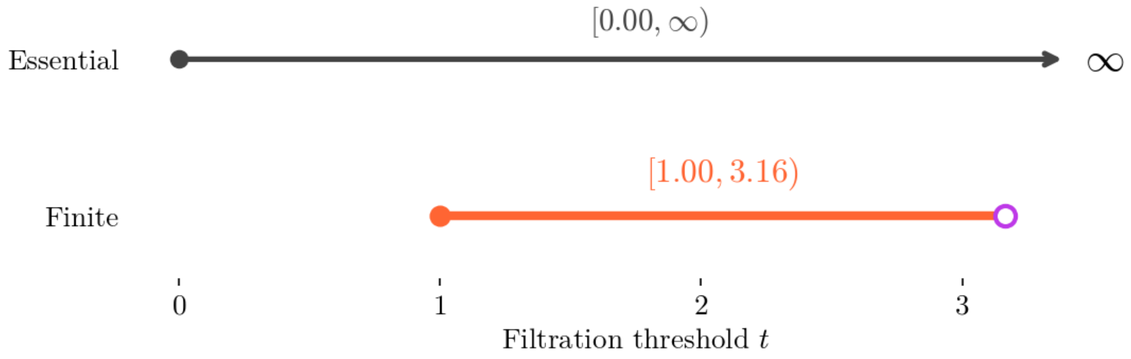
Note: you can extend this to work with general filtrations, by making them simplex-wise choosing a total order of the simplices

In that case, the derivative should be considered as a *subgradient*, and choosing a total order selects an element in the subgradient

What the optimization of the gradient yields

An example





Consider the following filtration over the point cloud:

$$P = (p_0, p_1) \subset \mathbb{R}^2$$

- Points enter the filtration according to their height
- The edge enters the filtration according to the distance between the points

$$\mathcal{L}(2, 0, 2; \mathrm{PD}_0) = (d - b)^2 = (\|p_0 - p_1\|^2 - y_1)^2, \quad d < \infty$$

$$\frac{\partial \mathcal{L}}{\partial b} = -2(d - b), \quad \frac{\partial \mathcal{L}}{\partial d} = 2(d - b)$$

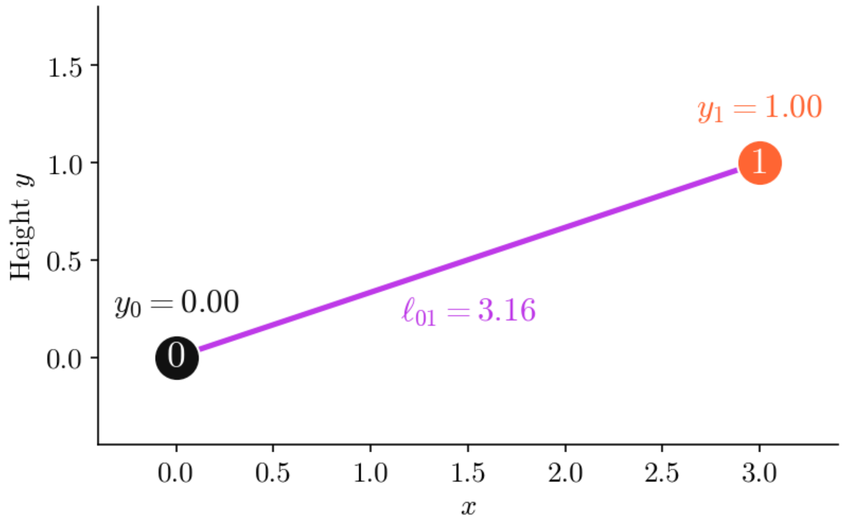
Optimization leads to a delaying the birth and anticipating the death

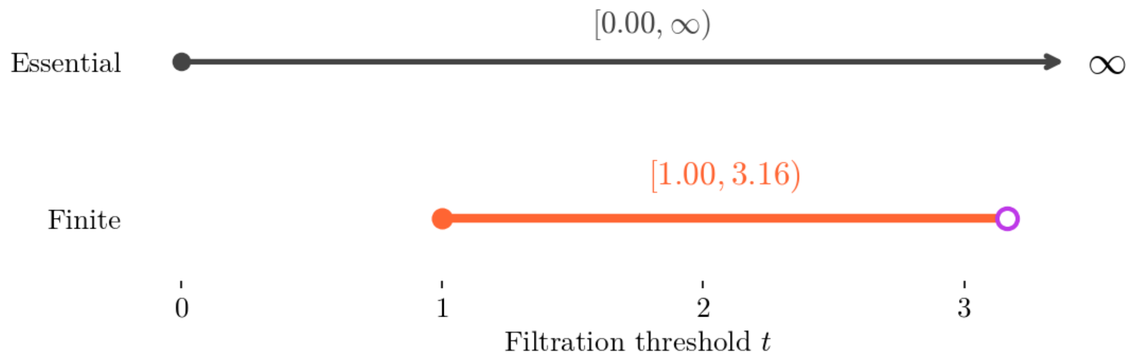
You can do the explicit computations in terms of the coordinates of the points, which yields...

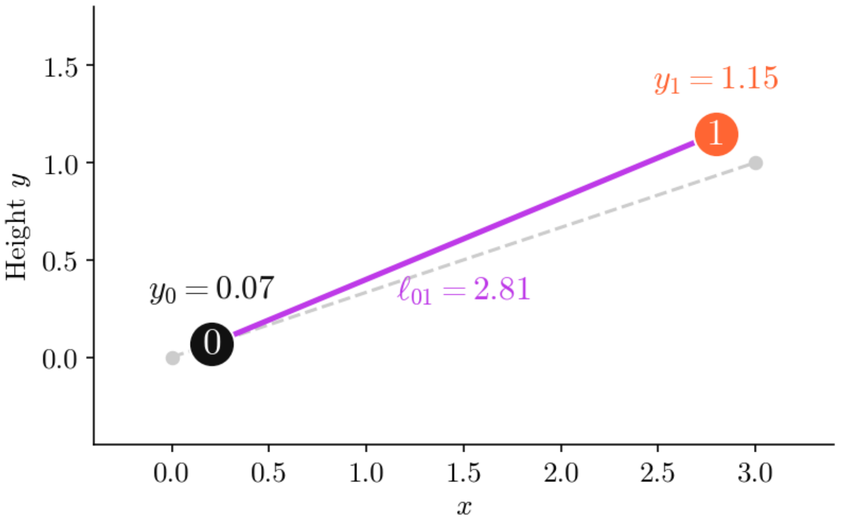
What the optimization of the gradient yields

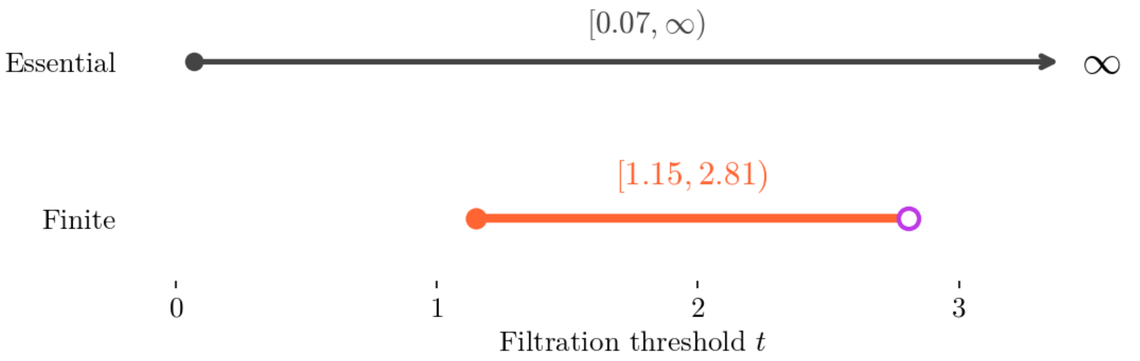
An example

$$P_{\text{new}} \mapsto P - 0.05\ \nabla_P \mathcal{L}$$





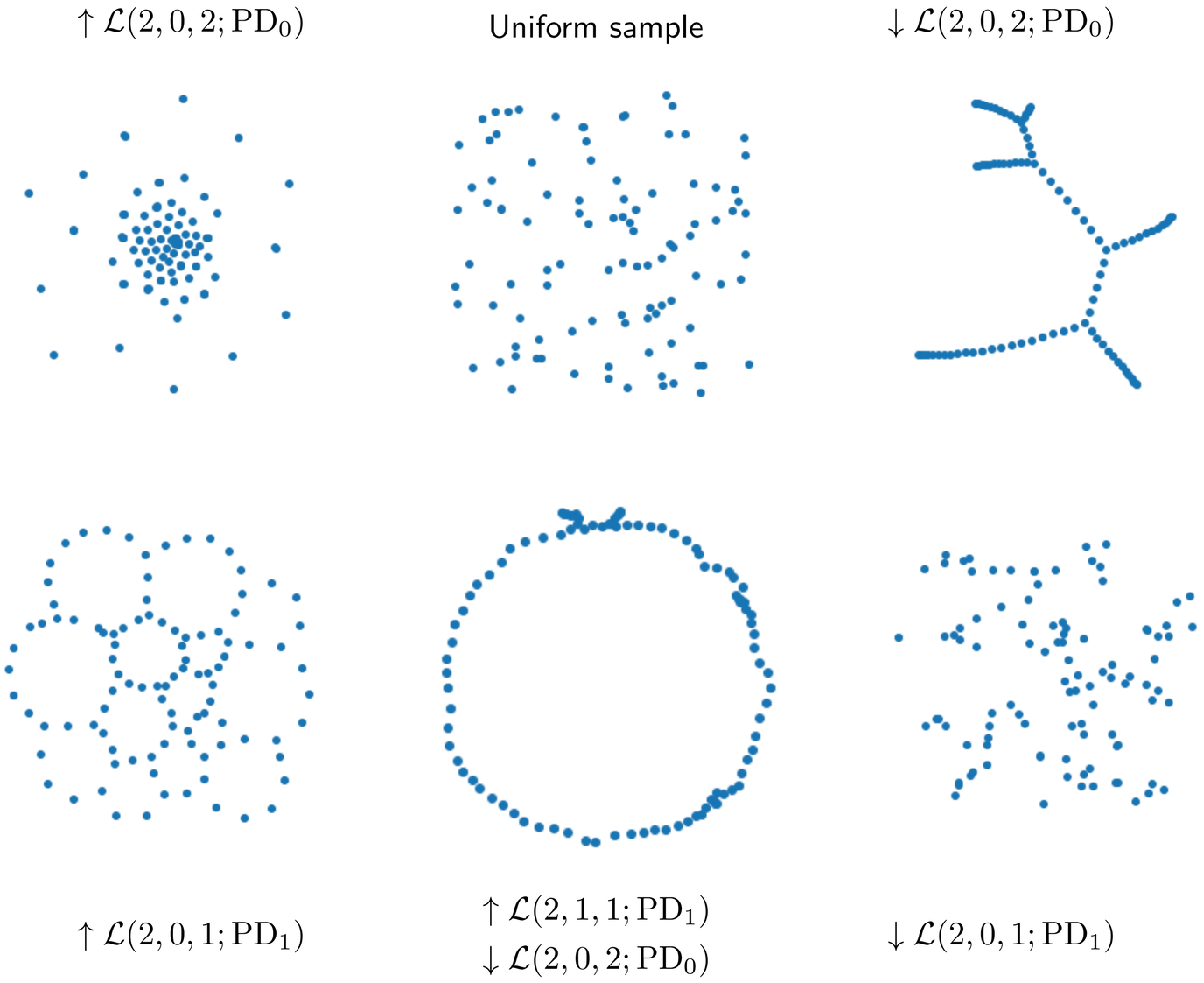




For point clouds

Figure from: Gabrielsson et *al.* (2020)

$$\mathcal{L}(2, 0, i_0; \mathrm{PD}_k) = \sum_{i=i_0}^{|\mathcal{I}_k|} |d_i - b_i|^2$$

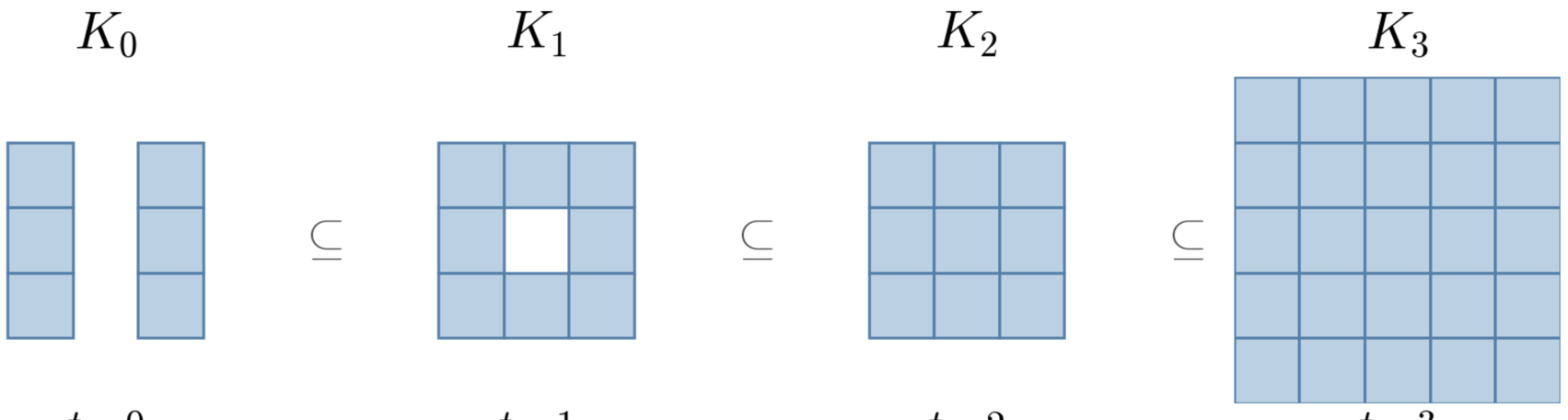


Gabrielsson, R. B., Nelson, B. J., Dwaraknath, A., and Skraba, P. (2020). A Topology Layer for Machine Learning. *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, 1553–1563.

Persistent homology for images

Pixel intensities

3	3	3	3	3
3	0	1	0	3
3	0	2	0	3
3	0	1	0	3
3	3	3	3	3



$$K_t = \bigcup_{I(p) \leq t} Q_p \qquad \text{(pixel squares and all their faces)}$$

Def. Sublevel set filtration

$$f : K \to \mathbb{R}, \qquad K_t = f^{-1}(-\infty, t]\{\sigma \in K : f(\sigma) \leq t\}$$

Topological loss

An application to noise reduction

Figure from: Gabrielsson et *al*. (2020)

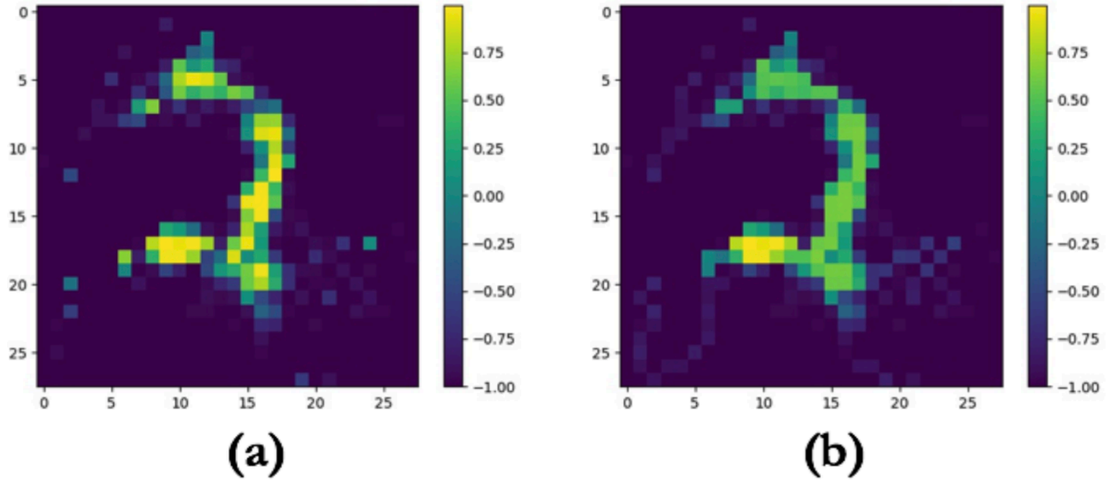


Figure 3: (a) A noisy image of the digit '2'. (b) after promoting a single local maximum using $\mathcal{L}(1, 0, 2; \mathrm{PD}_0)$

Gabrielsson, R. B., Nelson, B. J., Dwaraknath, A., and Skraba, P. (2020). A Topology Layer for Machine Learning. *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, 1553–1563.

Topological loss

An application to noise generative models

Figure from: Gabrielsson et *al.* (2020)

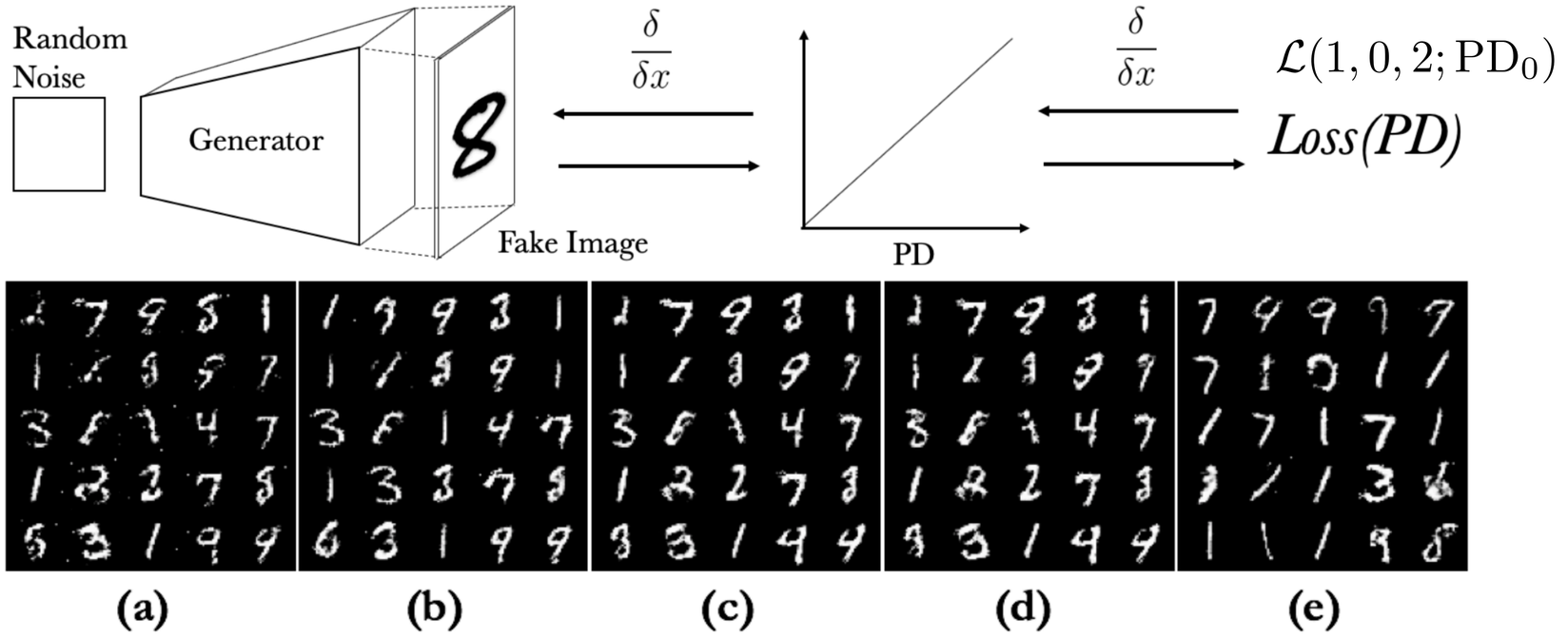


Figure 8: Top: Setup for computing topology loss and backpropogation. (a) Baseline-Generator. (b) Minimize the topology loss with respect to the latent space of Baseline-Generator. (c) Topology-Generator. (d) Train Baseline-Generator with topology-discriminator for 100 batch iterations. (e) Train Baseline-Generator in original GAN-setup for another 60,000 batch iterations.

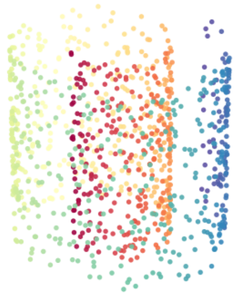
Gabrielsson, R. B., Nelson, B. J., Dwaraknath, A., and Skraba, P. (2020). A Topology Layer for Machine Learning. *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, 1553–1563.

Topological autoencoders

Session 2 — Geometric and topological methods in deep learning

Dimensionality reduction

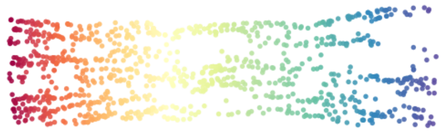
Input: $\mathbb{R}^3$



PCA



Isomap



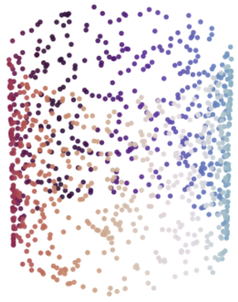
t-SNE



UMAP



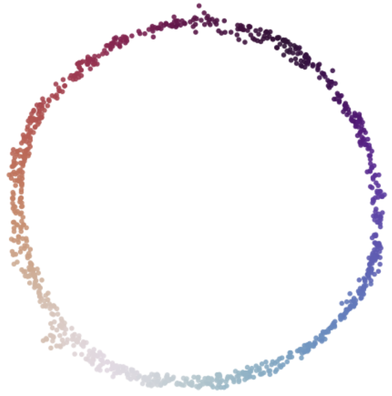
Input: $\mathbb{R}^3$



PCA



Isomap



t-SNE

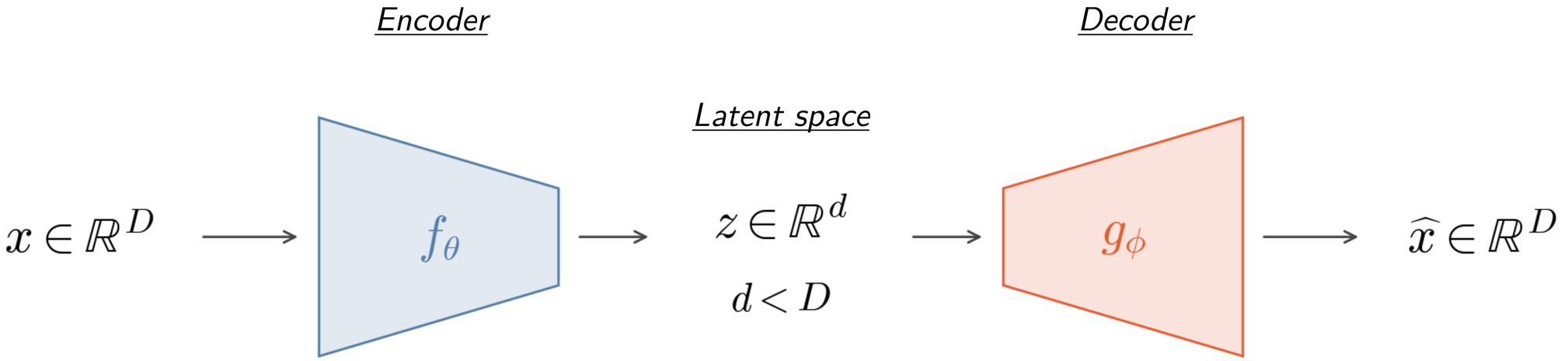


UMAP



Autoencoders

A brief introduction



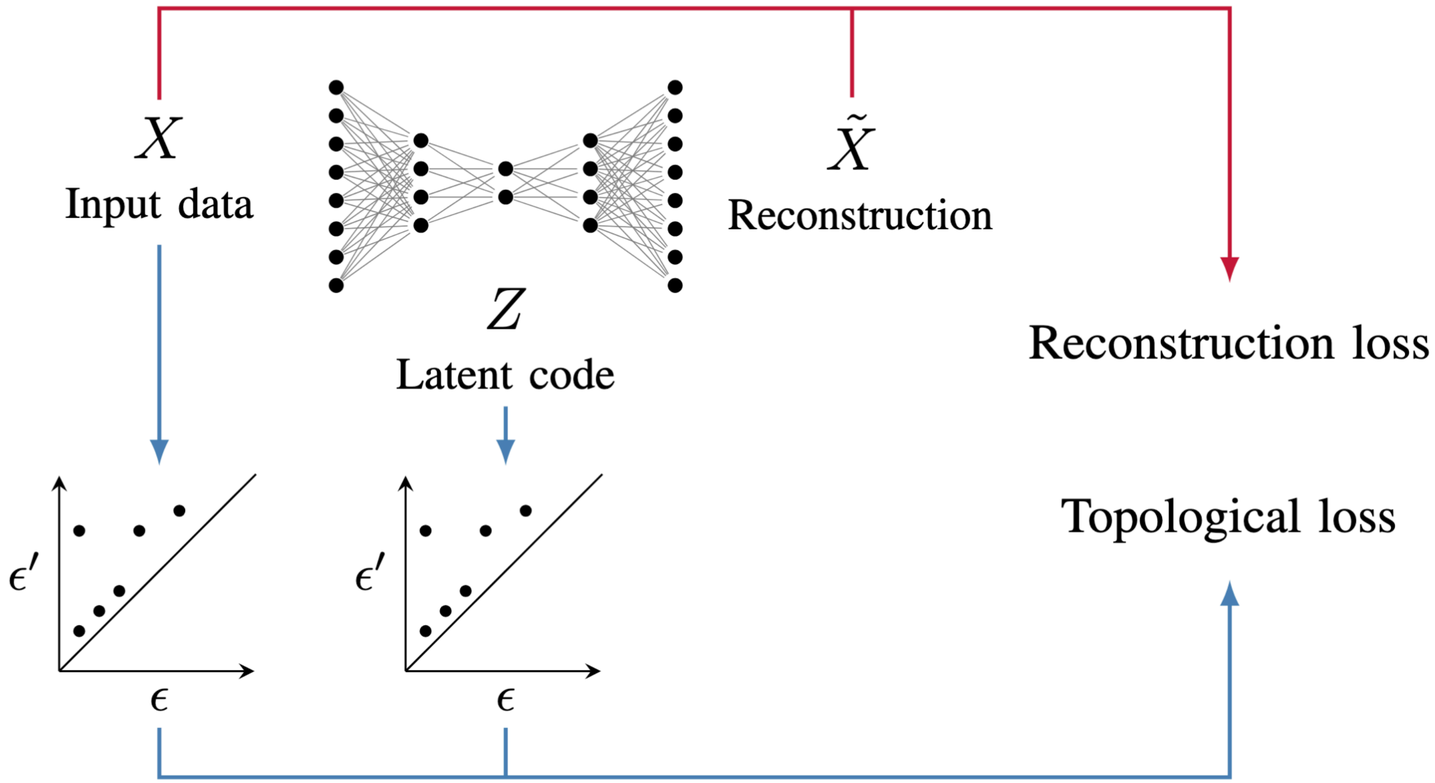
$$\mathcal{L}_{\text{rec}}(\theta, \phi) = \frac{1}{n} \sum_{i=1}^{n} \| x_i - g_\phi(f_\theta(x_i)) \|^2$$

Reconstruction loss

Topological loss for autoencoders

Overview

Figure from: Moor et *al.* (2020)



Moor, M., Horn, M., Rieck, B., and Borgwardt, K. (2020). Topological autoencoders. In *International conference on machine learning* (pp. 7045-7054). PMLR.

Topological loss for autoencoders

- Let X be a batch of input data, $Z = f_\theta(X)$ its latent representation (both are point clouds)
- Compute Vietoris—Rips persistent homology diagrams for both point clouds, giving:

$$(\mathrm{PD}^X, \pi^X), \qquad (\mathrm{PD}^Z, \pi^Z)$$

Persistence diagram

Matching between bars and critical cells

- You can obtain the bars by reading the distance matrices for the longest edges in each of the simplices

$$\mathrm{PD}^X \sim \mathbf{A}^X[\pi^X], \qquad \mathrm{PD}^Z \sim \mathbf{A}^Z[\pi^Z]$$

Matrix of pairwise distances

- Because both point clouds have the same number of points we can also select distances from X (resp. Z) using the matchings π^X (resp. π^Z)

$$\mathcal{L}_{X\to Z} = \frac{1}{2}\left\|\mathbf{A}^X[\pi^X] - \mathbf{A}^Z[\pi^X]\right\|^2, \qquad \mathcal{L}_{Z\to X} = \frac{1}{2}\left\|\mathbf{A}^Z[\pi^Z] - \mathbf{A}^X[\pi^Z]\right\|^2$$

Moor, M., Horn, M., Rieck, B., and Borgwardt, K. (2020). Topological autoencoders. In *International conference on machine learning* (pp. 7045-7054). PMLR.

Topological autoencoders

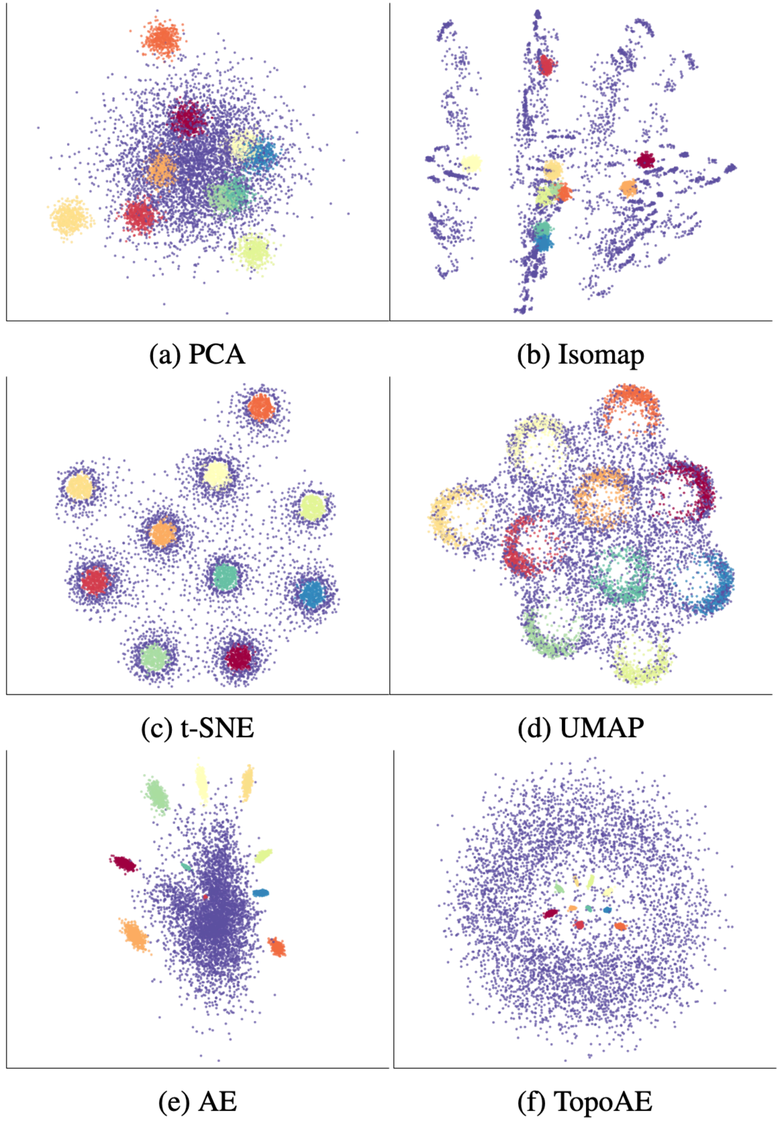
An application to synthetic data

- Create a synthetic dataset of spheres of dimension 100 in $\mathbb{R}^{101}$
- 10 spheres of radius $r = 5$, and random small displacements of their centers to the origin
- 1 extra sphere of radius $5r$ enclosing all of them
- Use point clouds from this model to train a topological auto encoder, and compare the latent representation of a point cloud to common dimensionality reduction techniques

The topological auto encoder is the method that best captures the nesting structure of the data!

The authors also present embedding quality measures for MNIST, FMNIST and CIFAR that favor the method

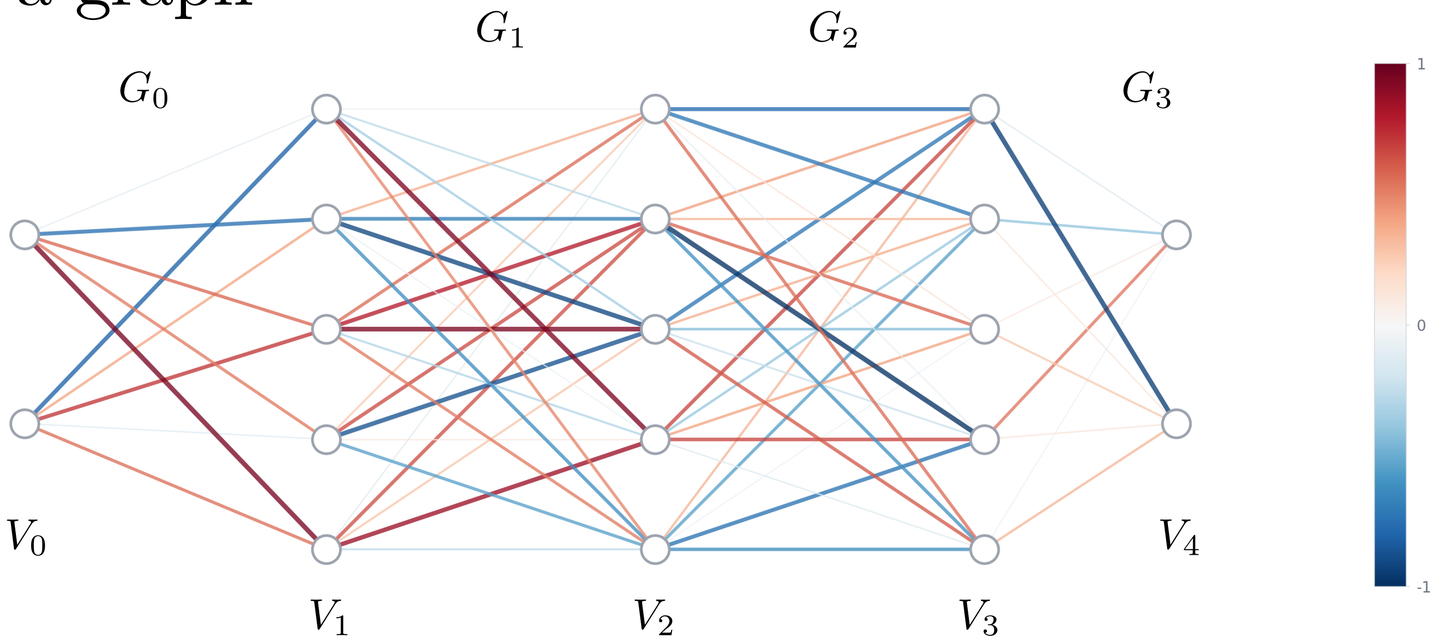
Figure from: Moor et *al.* (2020)



Neural persistence

Session 2 — Geometric and topological methods in deep learning

Seeing an MLP as a graph



We can see an MLP as a *stratified graph*: $G = (V, E)$

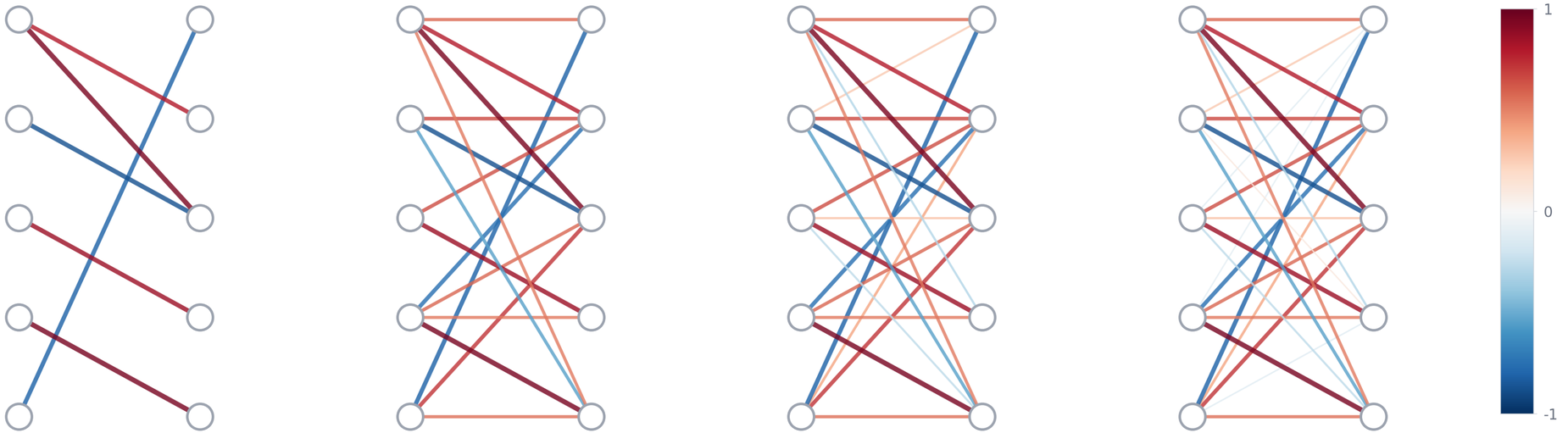
$$V = V_0 \sqcup V_1 \sqcup \cdots \sqcup V_L \qquad (u, v) \in E \iff \exists k \in \{0, \ldots, L\},\ u \in V_k \wedge v \in V_{k+1}$$

With a weight function on the edges: $\varphi : E \to \mathcal{W}$

$$G_k = (V_k \sqcup V_{k+1}, E \cap \{(u, v) \in V_k \times V_{k+1}\})$$

Rieck, Bastian, Matteo Togninalli, Christian Bock, et al. Neural Persistence: A Complexity Measure for Deep Neural Networks Using Algebraic Topology. May 2023, 25 p.

Defining the filtration



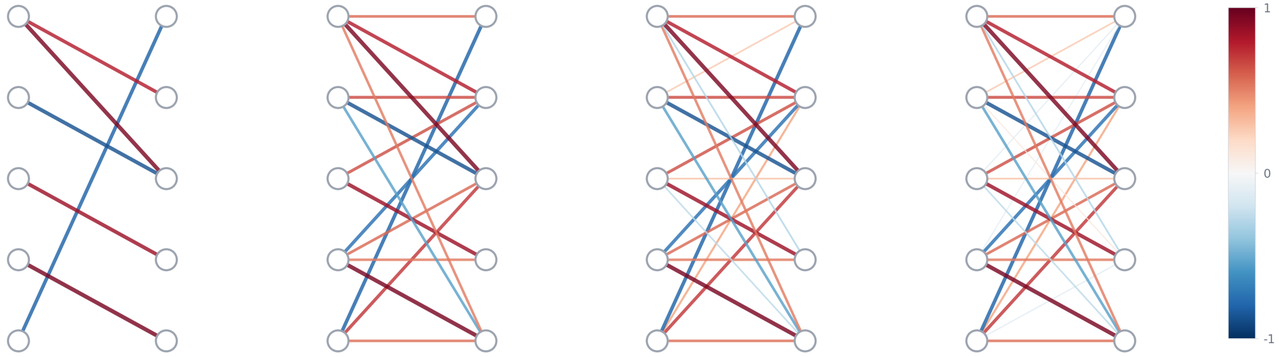
$$G_1^{(1)} \subseteq G_1^{(2)} \subseteq G_1^{(3)} \subseteq G_1^{(4)}$$

$$|w| > 0.75 \qquad |w| > 0.5 \qquad |w| > 0.25 \qquad |w| > 0$$

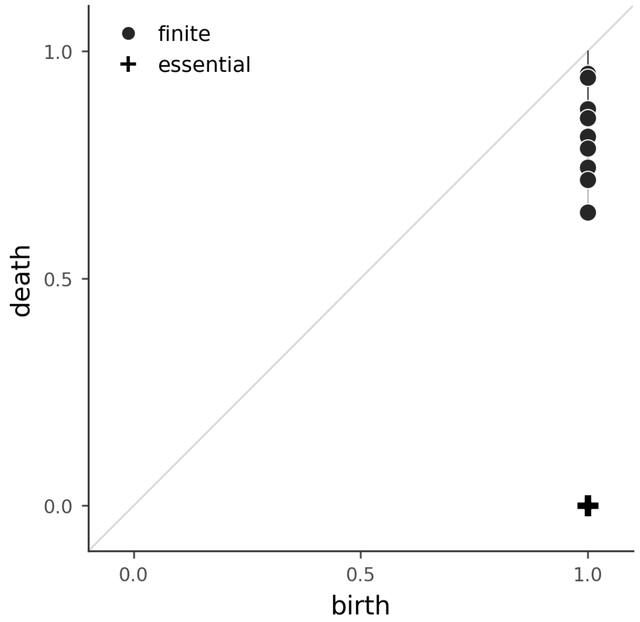
Rieck, Bastian, Matteo Togninalli, Christian Bock, et al. Neural Persistence: A Complexity Measure for Deep Neural Networks Using Algebraic Topology. May 2023, 25 p.

Persistent homology

Filtration for G_k



Persistence diagram PD_0^k



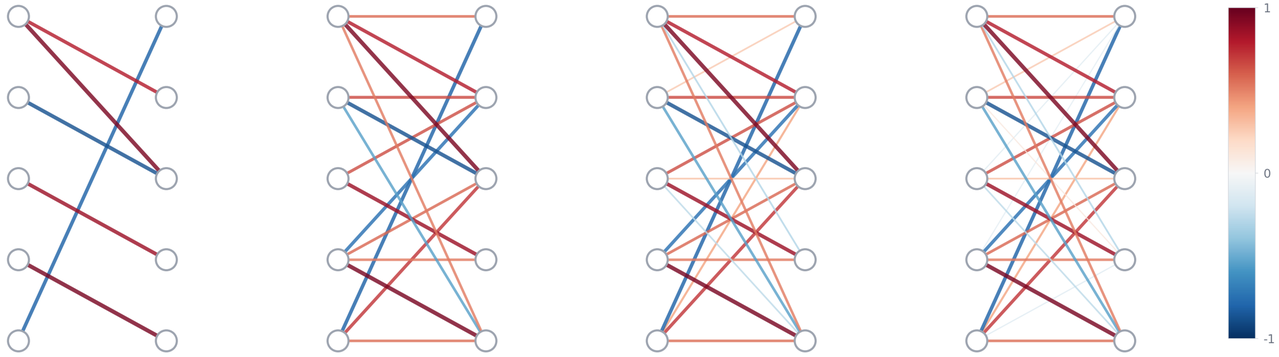
- Compute 0-dimensional PH
- All points are born at $w = 1$, and die for $w < 1$ as we add edges,
- Hence, the points in the diagram appear below the diagonal
- There is one essential component that stays alive until the end of the filtration

Rieck, Bastian, Matteo Togninalli, Christian Bock, et al. Neural Persistence: A Complexity Measure for Deep Neural Networks Using Algebraic Topology. May 2023, 25 p.

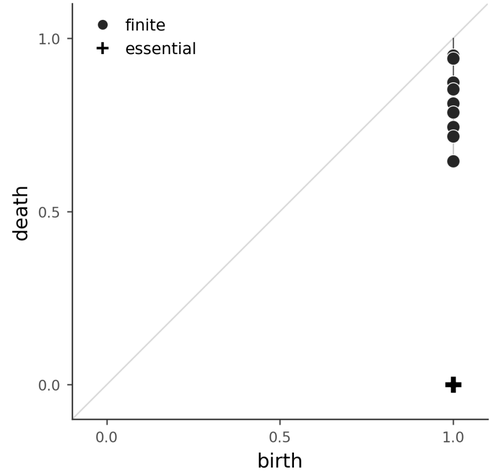
Neural persistence

For one layer

Filtration for G_k



Persistence diagram $\mathcal{D}_k$



Def. Neural persistence

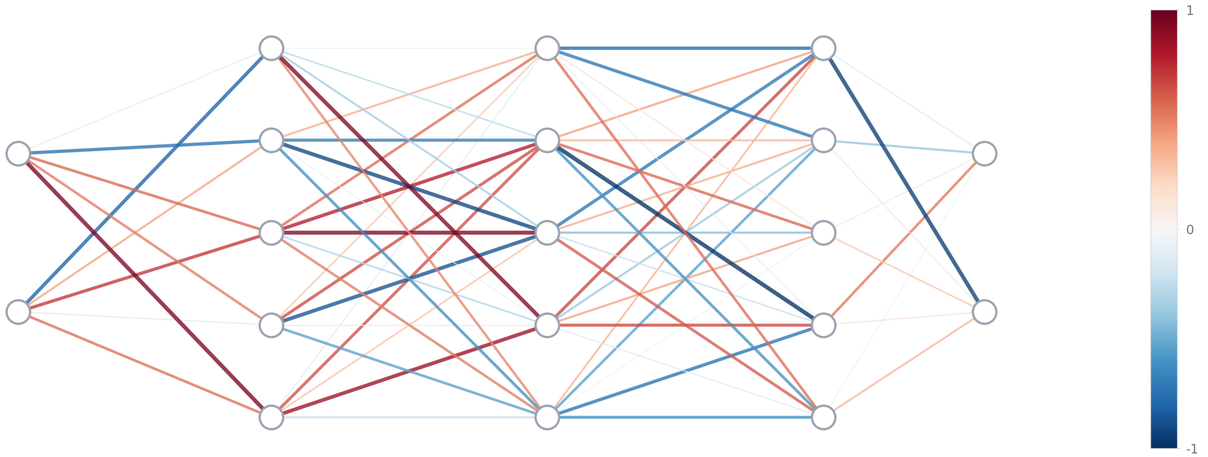
$$\mathrm{NP}(G_k) := \left\| \mathrm{PD}_0^k \right\| = \left(\sum_{(c,d) \in \mathrm{PD}_0^k} |d - c|^p \right)^{1/p}$$

$$0 \leq \mathrm{NP}(G_k) \leq \left(\max_{e \in G_k} w_e - \min_{e \in G_k} w_e \right) \left(|V_k \times V_{k+1}| - 1 \right)^{1/p}$$

Rieck, Bastian, Matteo Togninalli, Christian Bock, et al. Neural Persistence: A Complexity Measure for Deep Neural Networks Using Algebraic Topology. May 2023, 25 p.

Neural persistence

For the whole layer



$$\mathrm{NP}(G_k) := \left\| \mathrm{PD}_0^k \right\| = \left(\sum_{(c,d) \in \mathrm{PD}_0^k} |d - c|^p \right)^{1/p} \qquad 0 \leq \mathrm{NP}(G_k) \leq \mathrm{NP}(G_k)^+$$

$$\mathrm{NP}(G) := \frac{1}{L} \sum_{l=1}^{L} \frac{\mathrm{NP}(G_k)}{\mathrm{NP}(G_k)^+}$$

Rieck, Bastian, Matteo Togninalli, Christian Bock, et al. Neural Persistence: A Complexity Measure for Deep Neural Networks Using Algebraic Topology. May 2023, 25 p.

Neural persistence

Captures model complexity

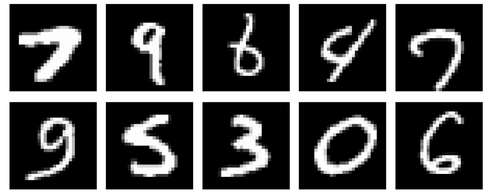
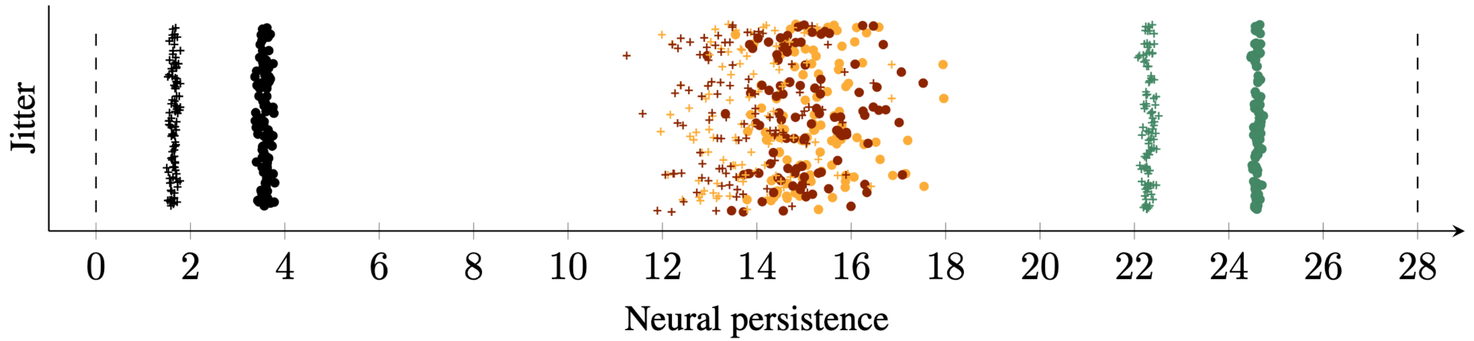


Figure from: Rieck et *al.* (2019)



$$\mathrm{NP}(G) := \frac{1}{L}\sum_{l=1}^{L}\frac{\mathrm{NP}(G_k)}{\mathrm{NP}(G_k)^+}$$

- Trained, converging models
- Trained, diverging models
- Gaussian weights
- Uniform weights

Rieck, Bastian, Matteo Togninalli, Christian Bock, et al. Neural Persistence: A Complexity Measure for Deep Neural Networks Using Algebraic Topology. May 2023, 25 p.

Neural persistence

Good training practices increase neural persistence

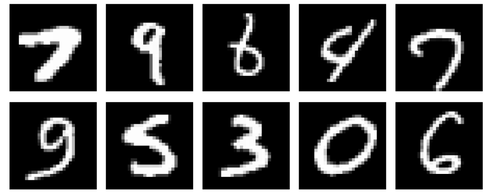
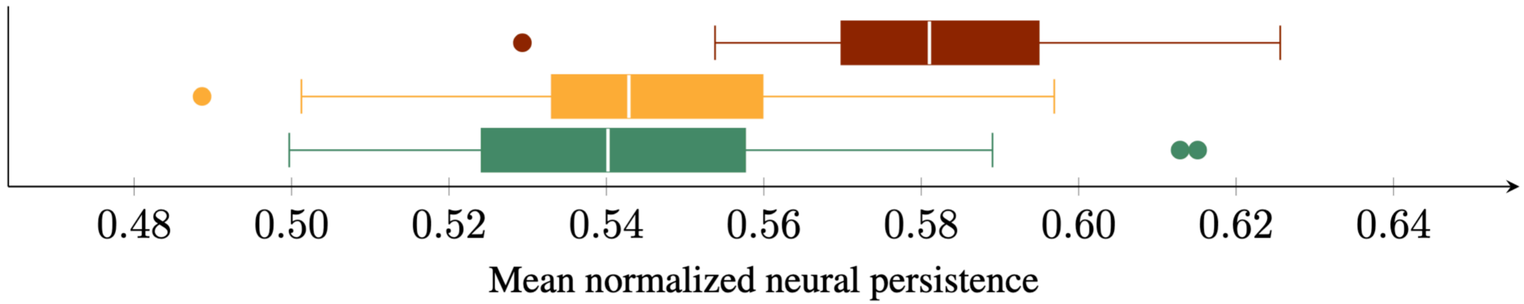


Figure from: Rieck et *al.* (2019)



$$\mathrm{NP}(G) := \frac{1}{L}\sum_{l=1}^{L}\frac{\mathrm{NP}(G_k)}{\mathrm{NP}(G_k)^+}$$

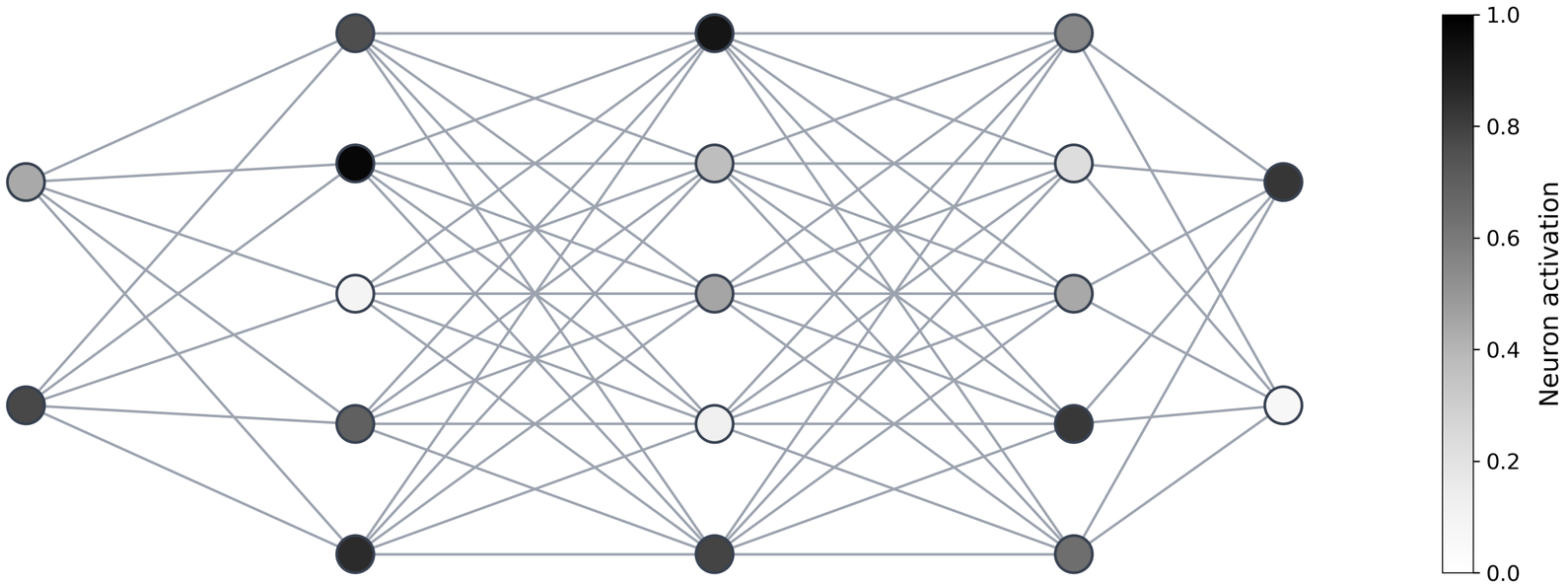
- No modification
- Batch normalization
- 50% dropout

Rieck, Bastian, Matteo Togninalli, Christian Bock, et al. Neural Persistence: A Complexity Measure for Deep Neural Networks Using Algebraic Topology. May 2023, 25 p.

PH to predict the generalization gap

Session 2 — Geometric and topological methods in deep learning

A PH construction to predict generalization



Training data

$$\mathcal{Z} = \{(x_i, y_i) : 1 \leq i \leq n\}$$

Activation value at v of x

$$N_v(x)$$

Activation vectors

$$A_v(\mathcal{Z}) = (N_v(x))_{(x,y)\in\mathcal{Z}}$$

$$A_N(\mathcal{Z}) = \{A_v(\mathcal{Z}) : v \in N\}$$

Correlation distance

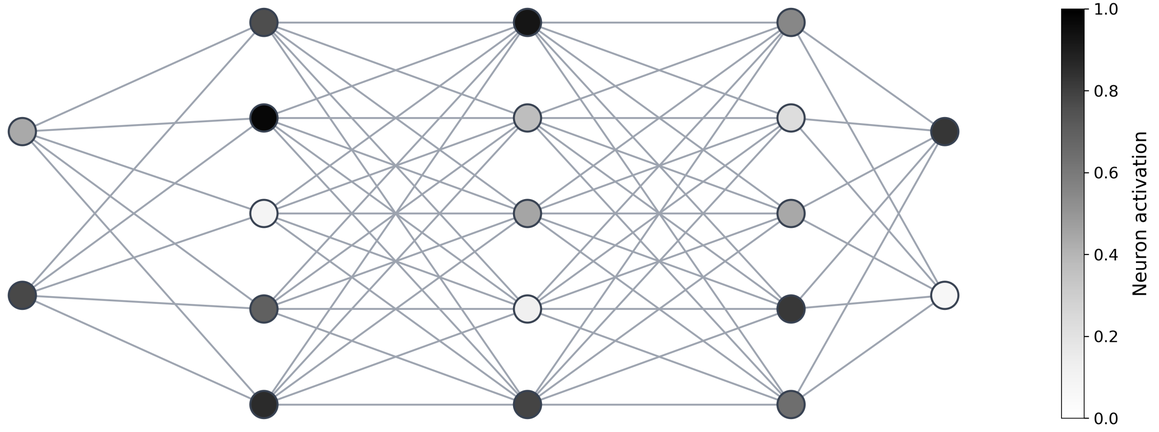
$$d(v_i, v_j) = 1 - |\mathrm{corr}(A_{v_i}(\mathcal{Z}), A_{v_j}(\mathcal{Z}))|$$

Pearson correlation coefficient

Ballester, R., Clemente, X. A., Casacuberta, C., Madadi, M., Corneanu, C. A., and Escalera, S. (2024). Predicting the generalization gap in neural networks using topological data analysis. *Neurocomputing*, *596*, 127787.

A PH construction to predict generalization

Networks coming from the Predicting Generalization In Deep Learning (PGDL) competition



Activation vectors

$$A_v(\mathcal{Z}) = (N_v(x))_{(x,y)\in\mathcal{Z}}$$

$$A_N(\mathcal{Z}) = \{A_v(\mathcal{Z}) : v \in N\}$$

$$|A_N(\mathcal{Z})| = |\mathcal{Z}| \times |V|$$

Computing PH for the whole point clouds becomes unfeasible

Instead consider subsamples:

$$\mathcal{Z}' \subset \mathcal{Z}, \quad |\mathcal{Z}'| = 2000$$

Randomly

$$N' \subset N, \quad |N'| = 3000$$

According to an *importance score*

Correlation distance

$$d(v_i, v_j) = 1 - |\mathrm{corr}(A_{v_i}(\mathcal{Z}), A_{v_j}(\mathcal{Z}))|$$

Pearson correlation coefficient

Repeat 20 times, compute statistical summaries of the diagrams, and build bootstrap samples

Jiang, Yiding, Pierre Foret, Scott Yak, et al. NeurIPS 2020 Competition: Predicting Generalization in Deep Learning. *CoRR* abs/2012.07976 (2020).

Ballester, R., Clemente, X. A., Casacuberta, C., Madadi, M., Corneanu, C. A., and Escalera, S. (2024). Predicting the generalization gap in neural networks using topological data analysis. *Neurocomputing*, *596*, 127787.

A PH construction to predict generalization

Results

$$R^2(y, \hat{y}) = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \overline{y}_i)^2}$$

Table 1
Top three combinations of persistence summaries per task according to their respective mean of R^2 test values in the 10 experiments of the 5 × 2-fold cross-validation statistical test. **ASD:** Average and standard deviation of births and deaths. **ASDSQ:** Average and standard deviation of births and deaths, concatenated with the corresponding squared values; see Section 4.1.

Task 1		
Top TDA summaries	Best dim	R^2 score
ASDSQ	0 and 1	0.5601 ± 0.13
ASDSQ	1	0.4321 ± 0.12
ASD	1	0.3720 ± 0.14
Task 2		
Top TDA summaries	Best dim	R^2 score
ASD	1	0.9337 ± 0.01
ASD	0 and 1	0.9198 ± 0.02
ASDSQ	1	0.9166 ± 0.03

Ballester, R., Clemente, X. A., Casacuberta, C., Madadi, M., Corneanu, C. A., and Escalera, S. (2024). Predicting the generalization gap in neural networks using topological data analysis. *Neurocomputing*, *596*, 127787.

A PH construction to predict generalization

Results

Table 2
Comparison of our best performing summaries with state of the art: Average and standard deviation of R^2 scores for Task 1 and Task 2 computed from linear models trained in the ten cases of the 5×2-fold cross-validation.

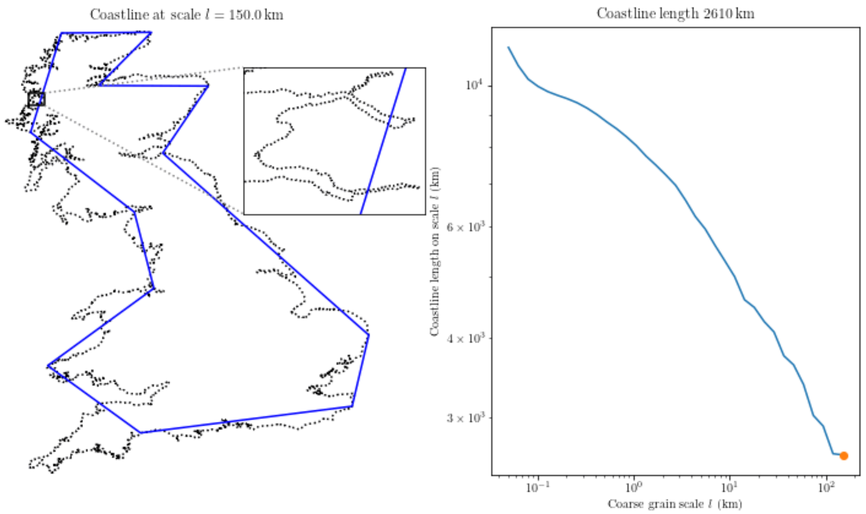
	Task 1	Task 2
Interpex	-0.0518 ± 0.06	0.9500 ± 0.01
Always Generalize	0.9715 ± 0.01	0.8893 ± 0.02
BrAIn	0.4520 ± 0.08	0.7180 ± 0.04
Ours	0.5601 ± 0.13	0.9337 ± 0.01

Ballester, R., Clemente, X. A., Casacuberta, C., Madadi, M., Corneanu, C. A., and Escalera, S. (2024). Predicting the generalization gap in neural networks using topological data analysis. *Neurocomputing*, *596*, 127787.

PH dimension and generalization

Session 2 — Geometric and topological methods in deep learning

Fractals

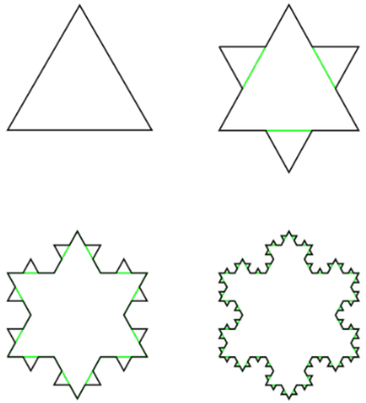


By Tveness - Own work, CC0, https://commons.wikimedia.org/w/index.php?curid=113571720

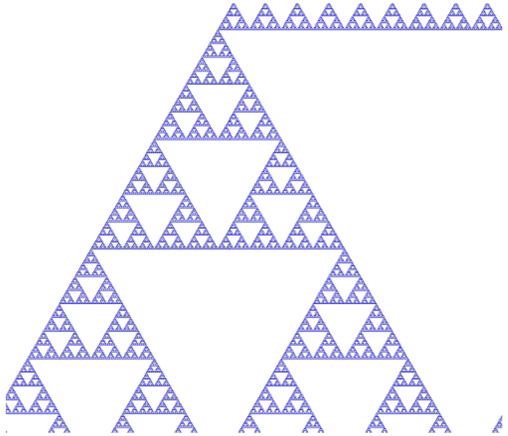
The coastline paradox

Self-similar shapes

Fractals can be seen as shapes that are *rough* when we zoom in



Von Koch snowflake, from Wikipedia



Zooming in the Sierpinski triangle, from Wikipedia

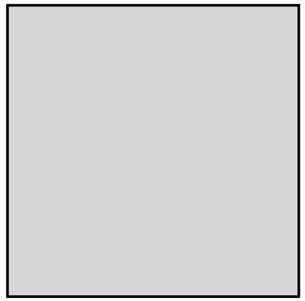
How can we define a notion of dimension that captures this *roughness?*

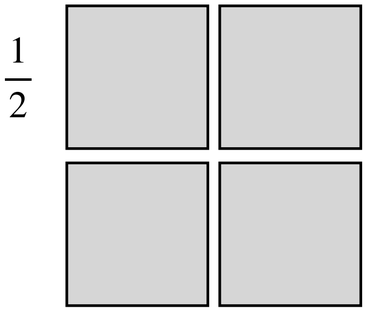
	Line	**Square**	**Cube**	**Sierpinski**
Scale factor	$s = \frac{1}{2}$			
Measure of the scaled-down shape	$\mu(L_s) = \frac{1}{2}$			



Intuition behind fractal dimension

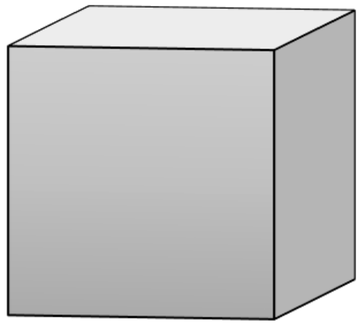
	Line	**Square**	**Cube**	**Sierpinski**
Scale factor	$s = \frac{1}{2}$	$s = \frac{1}{2}$		
Measure of the scaled-down shape	$\mu(L_s) = \frac{1}{2}$	$\mu(L_s) = \frac{1}{4}$		

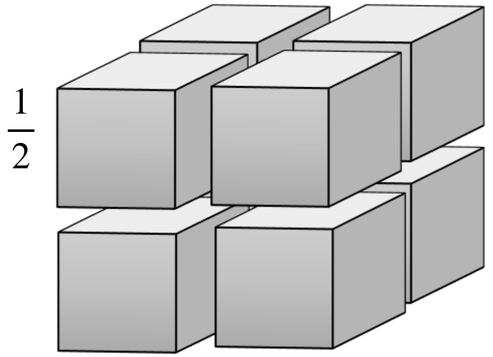




Intuition behind fractal dimension

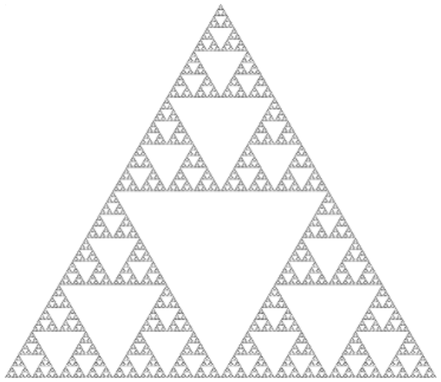
	Line	**Square**	**Cube**	**Sierpinski**
Scale factor	$s = \frac{1}{2}$	$s = \frac{1}{2}$	$s = \frac{1}{2}$	
Measure of the scaled-down shape	$\mu(L_s) = \frac{1}{2}$	$\mu(L_s) = \frac{1}{4}$	$\mu(L_s) = \frac{1}{8}$	

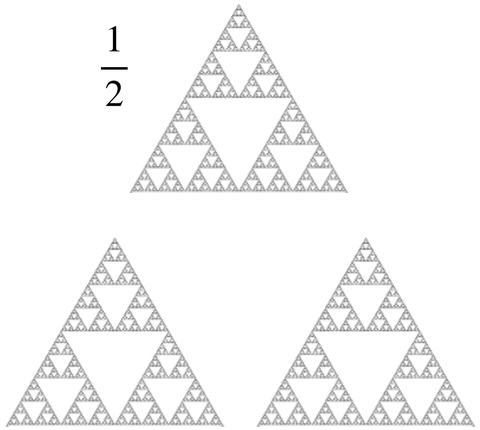




Intuition behind fractal dimension

	Line	**Square**	**Cube**	**Sierpinski**
Scale factor	$s = \frac{1}{2}$	$s = \frac{1}{2}$	$s = \frac{1}{2}$	$s = \frac{1}{2}$
Measure of the scaled-down shape	$\mu(L_s) = \frac{1}{2}$	$\mu(L_s) = \frac{1}{4}$	$\mu(L_s) = \frac{1}{8}$	$\mu(L_s) = \frac{1}{3}??$





Intuition behind fractal dimension

	Line	**Square**	**Cube**	**Sierpinski**
Scale factor	$s = \frac{1}{2}$	$s = \frac{1}{2}$	$s = \frac{1}{2}$	$s = \frac{1}{2}$
Measure of the scaled-down shape	$\mu(L_s) = \frac{1}{2}$	$\mu(L_s) = \left(\frac{1}{2}\right)^2$	$\mu(L_s) = \left(\frac{1}{2}\right)^3$	$\mu(L_s) = \left(\frac{1}{2}\right)^D$

Fractal dimension: D such that, if L is D-dimensional and we scale it down by s

$$\mu(L_s) = s^D$$

$$\left(\frac{1}{2}\right)^D = \frac{1}{3} \implies D = \log_2 3$$

Fractals have non-integer dimension

Box-counting dimension

A first definition

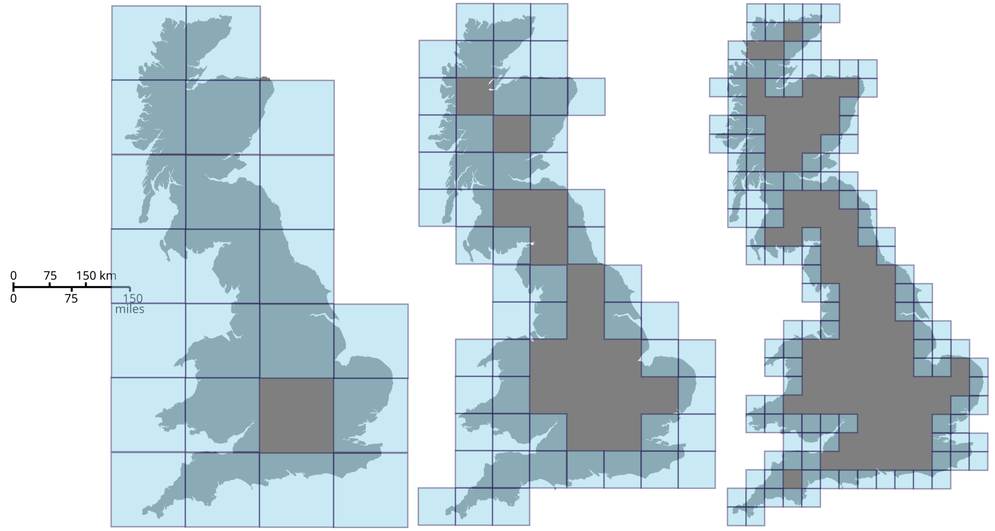
Metric space

balls to cover S

$$\dim_{\text{box}}(S) = \lim_{\epsilon \to 0} \frac{\log N_\epsilon}{\log(1/\epsilon)}$$

Radius

The limit might not exist



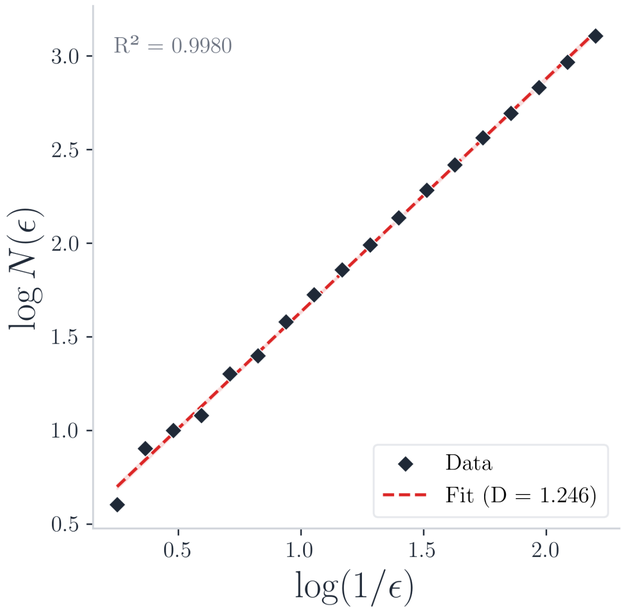
By Prokofiev - Own work, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=12042116

In practice, this means

$$N_\epsilon \approx C\epsilon^{-D}$$

$$\log N_\epsilon \approx \log C - d \log \epsilon$$

So we can estimate the box counting dimension interpolating a log-log plot



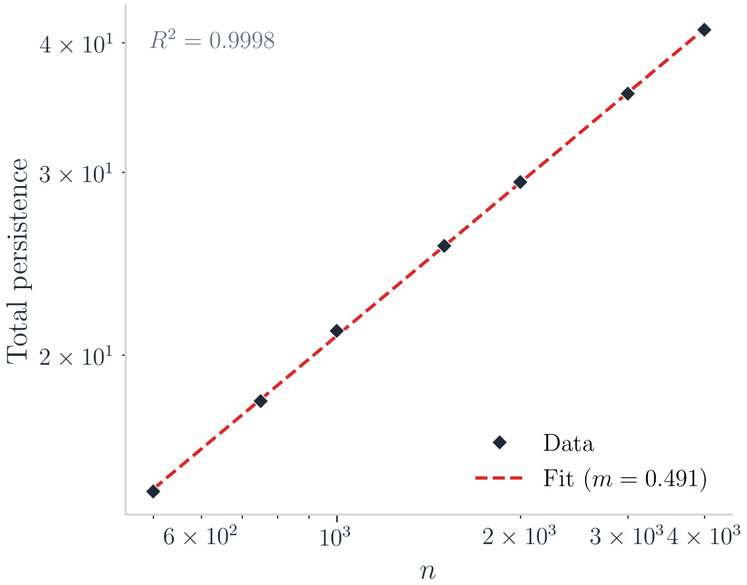
PH dimension

$$\mathbf{x} = \{x_1, \ldots, x_n\} \subset S \longrightarrow$$

Total persistence

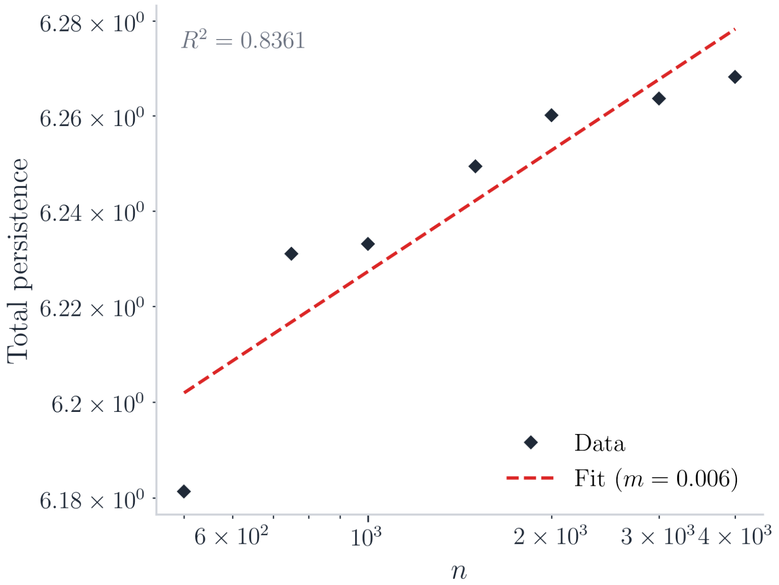
$$E_n(\mathbf{x}) = \sum_{(b,d) \in \mathrm{PD}_0(\mathbf{x})} |d - b|$$

$[0,1]^2$



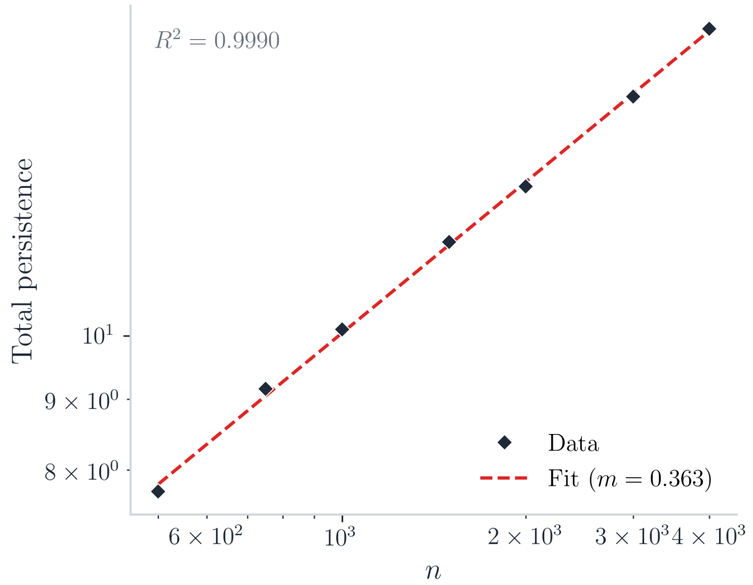
$$\frac{1}{1-m} \approx 1.996$$

Circle



$$\frac{1}{1-m} \approx 1.006$$

Sierpiński triangle



$$\frac{1}{1-m} \approx 1.571 \approx \frac{\log 3}{\log 2}$$

Schweinhart, Benjamin. Fractal Dimension and the Persistent Homology of Random Geometric Complexes. *Advances in Mathematics* 372 (October 2020): 107291

Jaquette, Jonathan, and Benjamin Schweinhart. Fractal Dimension Estimation with Persistent Homology: A Comparative Study. *Communications in Nonlinear Science and Numerical Simulation* 84 (May 2020): 105163.

Adams, Henry, Manuchehr Aminian, Elin Farnell, et al. *A Fractal Dimension for Measures via Persistent Homology*. Vol. 15. 2020.

PH dimension

You can formalize this to define a notion of *PH fractal dimension* and prove that is equivalent to the box-counting

$$\dim_{\mathrm{box}}(S) = \dim_{\mathrm{PH}}(S)$$

Kozma, Gady, Zvi Lotker, and Gideon Stupp. The Minimal Spanning Tree and the Upper Box Dimension. *Proceedings of the American Mathematical Society* 134, no. 4 (2006): 1183–87.

Schweinhart, Benjamin. Persistent Homology and the Upper Box Dimension. *Discrete & Computational Geometry* 65, no. 2 (2021): 331–64.

Bounding generalization with PH dimension

Optimization trajectory $\mathcal{W}_{\mathcal{Z}} = \{\theta_1, \ldots, \theta_T\} \subset \mathbb{R}^m$

Worst-case generalization error over the optimization trajectory $\max_{\theta \in \mathcal{W}_{\mathcal{Z}}} |\mathcal{R}(\theta) - \widehat{\mathcal{R}}(\theta, \mathcal{Z})|$

With probability $1 - \delta$

$$\max_{\theta \in \mathcal{W}_{\mathcal{Z}}} |\mathcal{R}(\theta) - \widehat{\mathcal{R}}(\theta, \mathcal{Z})| \leq C \sqrt{\frac{\dim_{\mathrm{PH}}(\mathcal{W}_{\mathcal{Z}}) + I(\mathcal{W}_{\mathcal{Z}}, \mathcal{Z}) + \log(1/\delta)}{n}}$$

- *Euclidean* distance in $\mathbb{R}^m$
- Data-dependent, *loss-based* pseudo metric:

$$\rho_{\mathcal{Z}}(\theta, \theta') = \frac{1}{n} \sum_{i=1}^{n} |\mathcal{L}(f_\theta(x_i), y_i) - \mathcal{L}(f_{\theta'}(x_i), y_i)|$$

Birdal, Tolga, Aaron Lou, Leonidas J. Guibas, and Umut Simsekli. "Intrinsic Dimension, Persistent Homology and Generalization in Neural Networks." *Advances in Neural Information Processing Systems* 34 (2021): 6776–89.

Dupuis, Benjamin, George Deligiannidis, and Umut Simsekli. "Generalization Bounds Using Data-Dependent Fractal Dimensions." *Proceedings of the 40th International Conference on Machine Learning*, July 3, 2023, 8922–68.

Bounding generalization with PH dimension

Experimental setup

- Train MLPs and CNNs with MNIST, CIFAR-10 (classification) and CHD (regression)
- Train with stochastic gradient descent, taking learning rates and batch sizes in a 6×6 grid

$$\theta_{k+1} = \theta_k - \eta \nabla \widetilde{\mathcal{R}}_k(\theta_k) \quad \text{with} \quad \nabla \widetilde{\mathcal{R}}_k(\theta) := \frac{1}{B} \sum_{i \in \widetilde{B}_k} \nabla \widehat{\mathcal{R}}(\theta, \mathcal{Z})$$

- Train until convergence and run 5,000 iterations more to compute PH dimension
- Plot against generalization gap of the trained model

Train accuracy - test accuracy

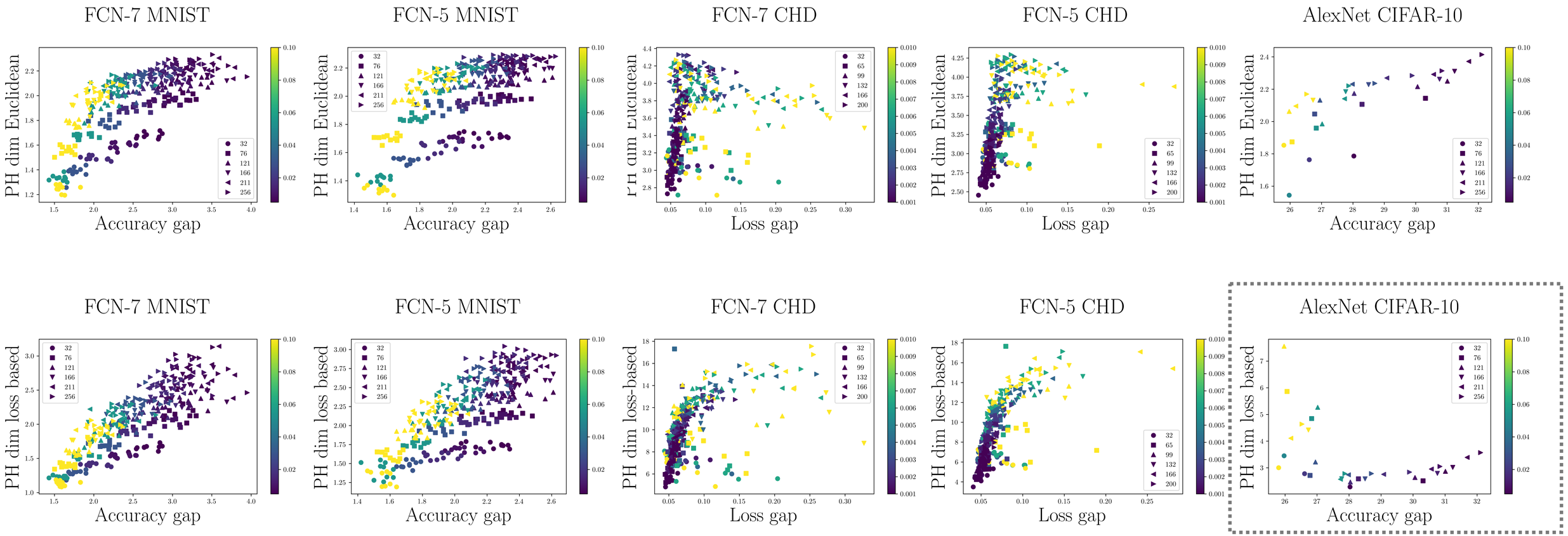
Want to observe a **positive correlation** between PH dimension and generalization

Dupuis, Benjamin, George Deligiannidis, and Umut Simsekli. "Generalization Bounds Using Data-Dependent Fractal Dimensions." *Proceedings of the 40th International Conference on Machine Learning*, July 3, 2023, 8922–68.

Tan, Charlie B., Inés García-Redondo, Qiquan Wang, Michael M. Bronstein, and Anthea Monod. "On the Limitations of Fractal Dimension as a Measure of Generalization." In *Advances in Neural Information Processing Systems*, vol. 37, edited by A. Globerson, L. Mackey, D. Belgrave, et al. Curran Associates, Inc., 2024.

Bounding generalization with PH dimension

Results



Tan, Charlie B., Inés García-Redondo, Qiquan Wang, Michael M. Bronstein, and Anthea Monod. "On the Limitations of Fractal Dimension as a Measure of Generalization." In *Advances in Neural Information Processing Systems*, vol. 37, edited by A. Globerson, L. Mackey, D. Belgrave, et al. Curran Associates, Inc., 2024.

Bounding generalization with PH dimension

Failure mode with adversarial initializations

Adversarial initialization (Liu et al., 2020):

- Randomize labels on training data
- Train model in randomized training
- Use optimized model as initialization for a regular training
- The resulting model will have **bad generalization** properties (big generalization gap)
- We expect these models to have **big PH dimension**

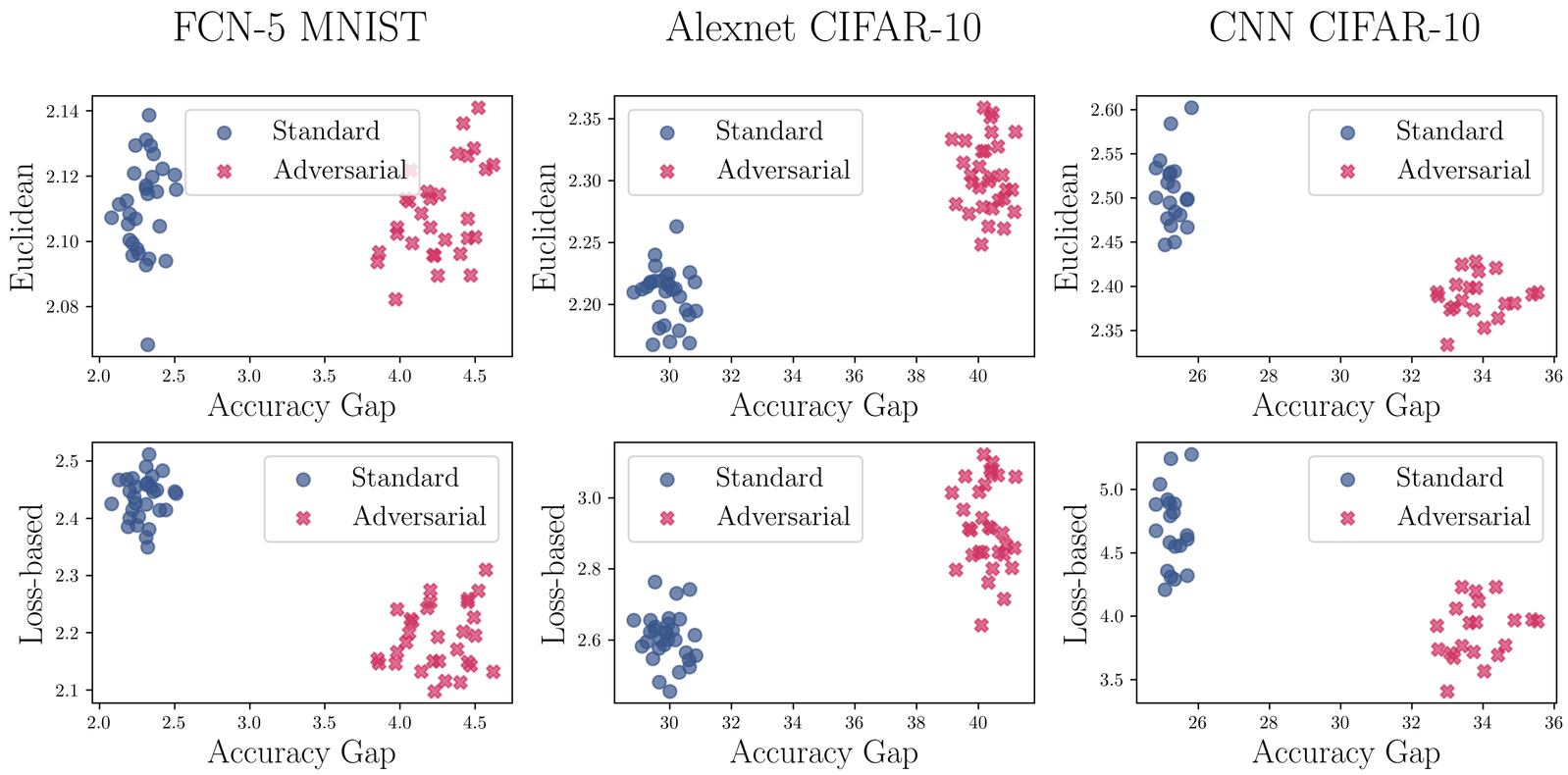
Standard initialization

- Models from standard, random initial points will tend to have better generalization properties
- We expect these have **smaller PH dimension**

Tan, Charlie B., Inés García-Redondo, Qiquan Wang, Michael M. Bronstein, and Anthea Monod. "On the Limitations of Fractal Dimension as a Measure of Generalization." In *Advances in Neural Information Processing Systems*, vol. 37, edited by A. Globerson, L. Mackey, D. Belgrave, et al. Curran Associates, Inc., 2024.

Bounding generalization with PH dimension

Failure mode with adversarial initializations



Tan, Charlie B., Inés García-Redondo, Qiquan Wang, Michael M. Bronstein, and Anthea Monod. "On the Limitations of Fractal Dimension as a Measure of Generalization." In *Advances in Neural Information Processing Systems*, vol. 37, edited by A. Globerson, L. Mackey, D. Belgrave, et al. Curran Associates, Inc., 2024.

Takeaway messages

The proposed correlation, in practice, seems to be heavily affected by the hyper parameter choices

This hypothesis is confirmed with partial correlation and conditional independence tests

The proposed correlation, fails to appear with adversarial initialization

What hypothesis from the theory are we breaking in this setting?

$$\max_{\theta \in \mathcal{W}_{\mathcal{Z}}} |\mathcal{R}(\theta) - \widehat{\mathcal{R}}(\theta, \mathcal{Z})| \leq C \sqrt{\frac{\dim_{\mathrm{PH}}(\mathcal{W}_{\mathcal{Z}}) + I(\mathcal{W}_{\mathcal{Z}}, \mathcal{Z}) + \log(1/\delta)}{n}}$$

New updated measures using *magnitude*

Andreeva, Rayna, Benjamin Dupuis, Rik Sarkar, Tolga Birdal, and Umut Şimşekli. "Topological Generalization Bounds for Discrete-Time Stochastic Optimization Algorithms." *Advances in Neural Information Processing Systems* 37 (December 2024): 4765–818.

To sum up

Today we have learnt how to learn from persistence diagrams and how to make the persistence computation *differentiable*, to define topological losses that take as input persistence diagrams.

We have also seen how to use PH to say something about generalization properties of neural networks, although not always what we expected...

Tomorrow we'll use PH to study the internal representations of data in DL models!

Thank you for your attention!



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